

(12) **United States Patent**
Richardson

(10) **Patent No.:** **US 12,384,208 B2**
(45) **Date of Patent:** **Aug. 12, 2025**

(54) **SYSTEM FOR TIRE INFLATION**

(56) **References Cited**

(71) Applicant: **Aperia Technologies, Inc.**, Hayward,
CA (US)

U.S. PATENT DOCUMENTS

(72) Inventor: **Brandon Richardson**, Burlingame, CA
(US)

789,024 A 5/1905 Hibbard et al.
1,358,524 A * 11/1920 Cooper B60C 23/129
152/421

(Continued)

(73) Assignee: **Aperia Technologies, Inc.**, Hayward,
CA (US)

FOREIGN PATENT DOCUMENTS

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

BR 102012023751 A2 8/2014
BR 112015017312 A2 7/2017

(Continued)

(21) Appl. No.: **18/510,463**

OTHER PUBLICATIONS

(22) Filed: **Nov. 15, 2023**

Richardson, Brandon , et al., "System and Method for Tire Inflator
Management", U.S. Appl. No. 18/538,645, filed Dec. 13, 2023.

(Continued)

(65) **Prior Publication Data**

US 2024/0083203 A1 Mar. 14, 2024

Primary Examiner — Dominick L Plakkootam

(74) *Attorney, Agent, or Firm* — Jeffrey Schox; Diana Lin

Related U.S. Application Data

(60) Continuation of application No. 18/097,479, filed on
Jan. 16, 2023, now Pat. No. 11,850,896, which is a
(Continued)

(51) **Int. Cl.**
B60C 23/10 (2006.01)
B60C 23/00 (2006.01)
(Continued)

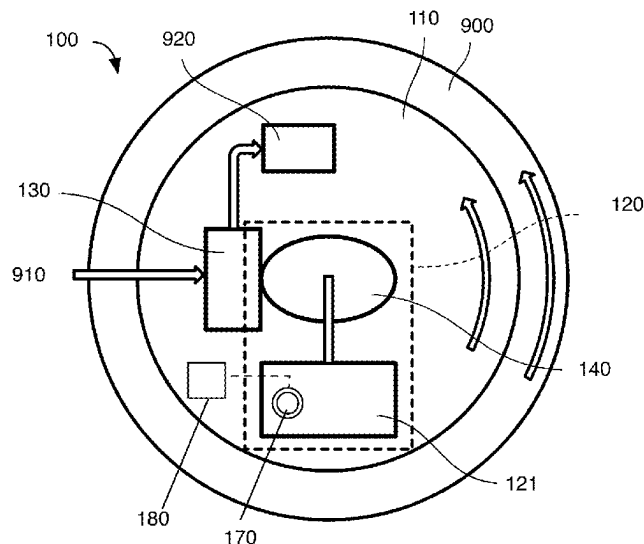
(52) **U.S. Cl.**
CPC **B60C 23/10** (2013.01); **B60C 23/00**
(2013.01); **B60C 23/12** (2013.01); **B60C**
23/127 (2020.05);
(Continued)

(58) **Field of Classification Search**
CPC B60C 23/10; B60C 23/00; B60C 23/12;
B60C 23/127; B60C 23/131;
(Continued)

(57) **ABSTRACT**

A system for tire inflation including a drive mechanism
defining a rotational axis, including an eccentric mass that
offsets a center of mass of the drive mechanism from the
rotational axis along a radial vector; a pump arranged
radially distal the rotational axis of the drive mechanism,
including a chamber defining a chamber lumen, and a
reciprocating element arranged at least partially within the
chamber lumen and translatable along a pump axis; a drive
coupler coupled between the drive mechanism at a first
position and the reciprocating element at a second position
fixed to the reciprocating element; a torque regulation
mechanism; and a controller, communicatively coupled to
the torque regulation mechanism; wherein the system is
operable between at least a first mode and a second mode by
the torque regulation mechanism in cooperation with the
controller.

20 Claims, 6 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/868,311, filed on Jul. 19, 2022, now Pat. No. 11,584,173, which is a continuation of application No. 17/061,313, filed on Oct. 1, 2020, now Pat. No. 11,453,258, which is a continuation-in-part of application No. 16/276,998, filed on Feb. 15, 2019, now Pat. No. 10,814,683, which is a division of application No. 15/696,816, filed on Sep. 6, 2017, now Pat. No. 10,245,908, said application No. 17/061,313 is a continuation-in-part of application No. 16/161,771, filed on Oct. 16, 2018, now Pat. No. 10,814,684, which is a continuation of application No. 15/280,737, filed on Sep. 29, 2016, now Pat. No. 10,144,254, which is a continuation-in-part of application No. 14/839,009, filed on Aug. 28, 2015, now abandoned, which is a continuation of application No. 14/198,967, filed on Mar. 6, 2014, now Pat. No. 9,039,386, which is a continuation of application No. 14/019,941, filed on Sep. 6, 2013, now Pat. No. 9,151,288, which is a continuation of application No. 13/797,826, filed on Mar. 12, 2013, now Pat. No. 9,039,392.

- (60) Provisional application No. 62/383,910, filed on Sep. 6, 2016, provisional application No. 62/519,061, filed on Jun. 13, 2017, provisional application No. 62/235,121, filed on Sep. 30, 2015.

(51) **Int. Cl.**

B60C 23/12 (2006.01)
F04B 35/01 (2006.01)
F04B 35/04 (2006.01)
F04B 39/00 (2006.01)

(52) **U.S. Cl.**

CPC **F04B 35/01** (2013.01); **F04B 35/04**
 (2013.01); **F04B 39/0005** (2013.01); **F04B**
2203/0207 (2013.01)

(58) **Field of Classification Search**

CPC B60C 23/004; F04B 35/01; F04B 35/04;
 F04B 39/0005; F04B 2203/0207; F04B
 39/121

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

1,448,248 A * 3/1923 Anthony B60C 23/129
 152/421
 1,484,914 A 2/1924 Thomas
 2,089,297 A 8/1937 Read et al.
 2,202,665 A 5/1940 Metz
 2,211,935 A 8/1940 Parker
 2,317,636 A 4/1943 Parker
 2,415,618 A 2/1947 West
 2,440,255 A 4/1948 Forrest
 2,496,691 A 2/1950 Berry
 2,899,907 A 8/1959 Becher
 2,962,119 A 11/1960 White
 3,012,820 A 12/1961 King
 3,047,339 A 7/1962 Hamer
 3,152,553 A 10/1964 Sverker
 3,175,507 A 3/1965 Sverker
 3,249,059 A 5/1966 Renn
 3,400,074 A 9/1968 Grenzi
 3,452,801 A * 7/1969 Fletcher B60C 23/137
 152/419
 3,511,294 A 5/1970 Beprestis et al.
 3,532,449 A 10/1970 Garton
 3,730,215 A 5/1973 Conery et al.

3,886,974 A 6/1975 Bjorklund
 3,981,633 A 9/1976 Wall
 4,018,579 A 4/1977 Hofmann
 4,030,300 A 6/1977 Thompson
 4,042,446 A 8/1977 Lawton
 4,095,923 A 6/1978 Cullis
 4,121,472 A 10/1978 Vural et al.
 4,157,530 A 6/1979 Merz
 4,180,985 A 1/1980 Northrup
 4,256,971 A 3/1981 Griffith
 4,291,235 A 9/1981 Bergey et al.
 4,370,894 A 2/1983 Stureson
 4,460,006 A 7/1984 Kolze
 4,536,668 A 8/1985 Boyer
 4,606,710 A 8/1986 Maguire
 4,637,152 A 1/1987 Roy
 4,651,792 A 3/1987 Taylor
 4,744,399 A 5/1988 Magnuson et al.
 4,768,542 A 9/1988 Morris
 4,807,487 A 2/1989 Seidl
 4,819,593 A 4/1989 Bruener et al.
 4,852,528 A 8/1989 Richeson et al.
 4,893,459 A 1/1990 Orlando
 4,895,199 A 1/1990 Magnuson et al.
 4,910,148 A 3/1990 Sorensen et al.
 4,922,984 A 5/1990 Dosjoub et al.
 5,018,797 A 5/1991 Takata
 5,052,456 A 10/1991 Dosjoub
 5,173,038 A 12/1992 Hopfensperger et al.
 5,201,968 A 4/1993 Renier
 5,293,919 A 3/1994 Olney et al.
 5,325,902 A 7/1994 Loewe et al.
 5,342,177 A 8/1994 Cheng
 5,370,711 A 12/1994 Audit et al.
 5,375,984 A 12/1994 Wehling
 5,388,470 A 2/1995 Marsh
 5,409,049 A 4/1995 Renier
 5,413,159 A 5/1995 Olney et al.
 5,429,927 A 7/1995 Afseth et al.
 5,468,129 A 11/1995 Sunden et al.
 5,482,447 A 1/1996 Sunden et al.
 5,495,879 A 3/1996 Cabestrero
 5,512,439 A 4/1996 Hornes et al.
 5,525,493 A 6/1996 Hornes et al.
 5,538,062 A 7/1996 Stech
 5,591,281 A * 1/1997 Loewe F04B 35/00
 417/233
 5,646,727 A 7/1997 Hammer et al.
 5,667,606 A 9/1997 Renier
 5,707,215 A * 1/1998 Olney B60C 23/133
 417/328
 5,759,820 A 6/1998 Hornes et al.
 5,846,354 A 12/1998 Winston et al.
 5,894,757 A 4/1999 Sully
 5,928,444 A * 7/1999 Loewe B60C 23/004
 152/418
 5,941,692 A 8/1999 Olney et al.
 5,941,696 A 8/1999 Fenstermacher et al.
 5,947,696 A * 9/1999 Baumgarten F04B 35/06
 417/233
 6,092,545 A 7/2000 Bedore et al.
 6,144,295 A 11/2000 Adams et al.
 6,267,450 B1 7/2001 Gamble
 6,360,768 B1 3/2002 Galler
 6,463,798 B2 10/2002 Niekerk et al.
 6,474,832 B2 11/2002 Murray
 6,482,592 B1 11/2002 Lundeberg et al.
 6,494,693 B1 12/2002 Sunden
 6,663,348 B2 12/2003 Schwarz et al.
 6,688,417 B2 2/2004 Hansell
 6,742,386 B1 6/2004 Larson
 6,744,356 B2 6/2004 Hamilton et al.
 6,787,233 B1 9/2004 Molteberg et al.
 6,814,547 B2 11/2004 Childers et al.
 6,871,683 B2 3/2005 Cobb
 6,984,702 B2 1/2006 Fonnum et al.
 6,986,913 B2 1/2006 Fonnum et al.
 6,994,136 B2 2/2006 Stanczak
 7,013,931 B2 3/2006 Toit

(56)

References Cited

U.S. PATENT DOCUMENTS

7,029,576 B2	4/2006	Siegfriedsen	10,422,660 B2	9/2019	Creech et al.
7,117,731 B2	10/2006	Hrabal	10,513,156 B2	12/2019	Hrabal
7,160,707 B2	1/2007	Fonnum et al.	10,576,794 B2	3/2020	Lee
7,173,124 B2	2/2007	Deggerdal et al.	10,625,544 B2	4/2020	Balistreri
7,217,762 B1	5/2007	Joergedal et al.	10,814,683 B2 *	10/2020	Richardson F04B 35/01
7,222,487 B1	5/2007	Hinkley	11,199,186 B2	12/2021	Slesinski et al.
7,225,845 B2	6/2007	Ellmann	11,209,289 B2	12/2021	Creech et al.
7,237,590 B2	7/2007	Loewe	11,353,137 B2	6/2022	Scopesi
7,255,323 B1	8/2007	Kadhim	11,370,189 B2	6/2022	Tsiberidis
7,302,837 B2	12/2007	Wendte	11,584,173 B2 *	2/2023	Richardson F04B 39/121
7,322,392 B2	1/2008	Hawes	11,642,920 B2	5/2023	Ciocia et al.
7,357,164 B2	4/2008	Loewe	2001/0000305 A1	4/2001	Murray
7,498,683 B2	3/2009	Landwehr	2002/0092346 A1	7/2002	Niekerk et al.
7,506,663 B2	3/2009	Thomas et al.	2003/0021693 A1	1/2003	Schwarz et al.
7,581,576 B2 *	9/2009	Nakano B60C 23/137	2003/0109657 A1	6/2003	Fonnum et al.
		417/221	2004/0007302 A1	1/2004	Hamilton et al.
7,607,465 B2	10/2009	Loewe	2004/0050760 A1	3/2004	Siegfriedsen
7,614,474 B2	11/2009	Yang	2004/0216827 A1	11/2004	Stanczak
7,625,189 B2	12/2009	Cheng	2005/0014001 A1	1/2005	Fonnum et al.
7,704,057 B2	4/2010	Malbec et al.	2005/0067207 A1	3/2005	Radtke et al.
7,748,422 B2	7/2010	Bol	2005/0126273 A1	6/2005	Hrabal
7,763,689 B2	7/2010	Fonnum et al.	2005/0147822 A1	7/2005	Fonnum et al.
7,784,487 B2	8/2010	Arnett	2005/0239068 A1	10/2005	Bosnes
7,784,513 B2 *	8/2010	Loewe B60C 23/043	2006/0055174 A1	3/2006	Landwehr
		152/418	2006/0102268 A1	5/2006	Loewe
7,810,582 B2	10/2010	Webb	2006/0118224 A1	6/2006	Ellmann
7,828,095 B2	11/2010	Murata et al.	2006/0169349 A1 *	8/2006	Nakano B60C 23/137
7,828,101 B2	11/2010	Radtke et al.			141/38
7,847,421 B2	12/2010	Gardner et al.	2007/0017595 A1	1/2007	Arno et al.
7,856,871 B2	12/2010	Mancosu et al.	2007/0018458 A1	1/2007	Martinez
7,927,170 B2	4/2011	Bickerton et al.	2007/0020130 A1	1/2007	Malbec et al.
7,931,061 B2	4/2011	Gonska et al.	2007/0040135 A1	2/2007	Dyer et al.
7,931,817 B2	4/2011	Bilski	2007/0068238 A1	3/2007	Wendt
7,975,789 B2	7/2011	Murata	2007/0107822 A1	5/2007	Loewe
7,985,340 B2	7/2011	Almaasbak et al.	2007/0148022 A1	6/2007	Cheng
7,989,614 B2	8/2011	Deggerdal et al.	2007/0151648 A1	7/2007	Loewe
7,989,975 B2	8/2011	Clement et al.	2007/0175831 A1	8/2007	Almaasbak et al.
8,004,104 B2	8/2011	Hench	2008/0003110 A1	1/2008	Isono
8,022,561 B2	9/2011	Ciglenec et al.	2008/0017256 A1	1/2008	Thomas et al.
8,038,987 B2	10/2011	Fonnum et al.	2008/0035399 A1	2/2008	Murata et al.
8,042,586 B2	10/2011	Losey et al.	2008/0135151 A1	6/2008	Loewe
8,083,503 B2	12/2011	Voltenburg et al.	2008/0149404 A1	6/2008	Yang
8,110,351 B2	2/2012	Bosnes	2008/0174120 A1	7/2008	Gardner et al.
8,113,254 B2	2/2012	Benedict	2008/0190535 A1	8/2008	Concu
8,186,402 B2	5/2012	Eigenbrode	2008/0247883 A1	10/2008	Yokomachi et al.
8,482,395 B2	7/2013	Rysenga et al.	2008/0289739 A1	11/2008	Bol
8,534,335 B2	9/2013	Benedict	2008/0300396 A1	12/2008	Deggerdal et al.
8,581,433 B2	11/2013	Sidenmark	2008/0308330 A1	12/2008	Murata
8,646,426 B2	2/2014	Finkenbiner	2009/0008942 A1	1/2009	Clement et al.
8,747,084 B2	6/2014	Richardson et al.	2009/0032158 A1	2/2009	Rudolf et al.
8,763,661 B2	7/2014	Richardson	2009/0048733 A1	2/2009	Isono
8,794,280 B2	8/2014	Flory et al.	2009/0087325 A1	4/2009	Voltenburg et al.
8,801,910 B2	8/2014	Bazant et al.	2009/0126997 A1	5/2009	Webb
8,955,566 B2	2/2015	Loewe	2009/0151835 A1	6/2009	Manning
9,039,386 B2	5/2015	Richardson et al.	2009/0188310 A1	7/2009	Mancosu et al.
9,039,392 B2	5/2015	Richardson et al.	2009/0206024 A1	8/2009	Bilski
9,074,595 B2	7/2015	Richardson	2009/0256532 A1	10/2009	Ciglenec et al.
9,145,887 B2	9/2015	Richardson	2009/0301575 A1	12/2009	Arnett
9,151,288 B2	10/2015	Richardson et al.	2009/0311941 A1	12/2009	Bickerton et al.
9,381,780 B2	7/2016	Hinque	2010/0102564 A1	4/2010	Hench
9,428,016 B2	8/2016	Keeney et al.	2010/0147429 A1	6/2010	Gonska et al.
9,434,218 B2	9/2016	Root	2010/0243121 A1	9/2010	Eigenbrode
9,597,932 B2	3/2017	Dieckmann et al.	2011/0018275 A1	1/2011	Sidenmark
9,682,599 B1 *	6/2017	Hinque B60C 23/004	2011/0061621 A1	3/2011	Finkenbiner
9,789,739 B1	10/2017	Hennig	2011/0146867 A1	6/2011	Benedict
9,994,081 B2	6/2018	Becker	2011/0146868 A1	6/2011	Losey et al.
10,052,834 B2	8/2018	Losey	2011/0203710 A1	8/2011	Hinojosa et al.
10,137,745 B2	11/2018	McClellan	2011/0308953 A1	12/2011	Bazant et al.
10,144,254 B2	12/2018	Richardson	2012/0020822 A1 *	1/2012	Richardson F04B 43/1238
10,166,821 B2	1/2019	Hennig			417/477.2
10,183,535 B1	1/2019	Chemali	2012/0073716 A1	3/2012	Benedict
10,189,320 B2 *	1/2019	Hinque B60C 23/126	2013/0068361 A1	3/2013	Flory et al.
10,239,364 B2	3/2019	Hennig	2013/0251552 A1	9/2013	Richardson
10,245,908 B2 *	4/2019	Richardson F04B 35/01	2013/0251553 A1 *	9/2013	Richardson B60C 23/00
10,358,005 B2	7/2019	McClellan			417/321
			2013/0276902 A1	10/2013	Medley et al.
			2014/0000755 A1 *	1/2014	Richardson F04B 35/01
					141/4

(56)

References Cited

U.S. PATENT DOCUMENTS

2014/0010680	A1 *	1/2014	Richardson	F04B 49/08
				417/321
2014/0023518	A1	1/2014	O'Brien et al.	
2014/0102621	A1	4/2014	Losey	
2014/0186195	A1 *	7/2014	Richardson	F04B 17/00
				417/221
2014/0260969	A1	9/2014	Richardson	
2015/0007920	A1	1/2015	Tanno	
2015/0059950	A1	3/2015	Hinque	
2015/0101723	A1	4/2015	Keeney et al.	
2015/0239309	A1	8/2015	Root	
2015/0360522	A1	12/2015	Dieckmann et al.	
2015/0367693	A1	12/2015	Richardson et al.	
2016/0008739	A1	1/2016	Richardson	
2016/0107491	A1 *	4/2016	Lin	F04B 27/005
				141/4
2016/0167465	A1	6/2016	Hinque	
2016/0288592	A1 *	10/2016	Becker	B60C 23/007
2017/0015147	A1 *	1/2017	McClellan	B60C 23/126
2017/0015156	A1 *	1/2017	McClellan	B60C 23/126
2017/0015159	A1	1/2017	Richardson	
2017/0036162	A1	2/2017	Hoeldrich	
2017/0072752	A1 *	3/2017	Lin	B60C 23/126
2017/0166017	A1	6/2017	Hinque	
2017/0166018	A1	6/2017	Hinque et al.	
2017/0299077	A1	10/2017	Jabcon et al.	
2018/0001718	A1	1/2018	Hennig	
2018/0022170	A1	1/2018	Hennig	
2018/0065429	A1	3/2018	Richardson	
2018/0072112	A1	3/2018	Lee	
2018/0104994	A1	4/2018	Lin et al.	
2018/0170125	A1	6/2018	Woodley	
2018/0178598	A1	6/2018	Woodley	
2018/0207575	A1	7/2018	Dunning et al.	
2018/0231395	A1	8/2018	Creech et al.	
2018/0250991	A1	9/2018	Hrabal	
2018/0326932	A1	11/2018	Chemali	
2019/0023085	A1	1/2019	Chemali	
2019/0063414	A1 *	2/2019	Slesinski	F04B 9/042
2019/0118593	A1	4/2019	Hrabal	
2019/0126693	A1	5/2019	Balistreri	
2019/0271407	A1	9/2019	Scopesi	
2019/0368897	A1	12/2019	Creech et al.	
2020/0070450	A1	3/2020	Tsiberidis	
2020/0130435	A1 *	4/2020	Root	B60C 23/127
2020/0130436	A1	4/2020	Root et al.	
2020/0130437	A1	4/2020	Root et al.	
2020/0134942	A1	4/2020	Root et al.	
2021/0016615	A1 *	1/2021	Richardson	B60C 23/004
2021/0061028	A1	3/2021	Da Deppo et al.	
2021/0125428	A1	4/2021	Tedesco et al.	
2022/0161611	A1	5/2022	Hrabal	
2022/0169084	A1 *	6/2022	Van Ginderen	B60C 23/127

FOREIGN PATENT DOCUMENTS

CA	2440255	A1	3/2005
CN	2202665	Y	7/1995
CN	2496691	Y	6/2002
CN	2635411	Y	8/2004
CN	101415946	A	4/2009
CN	101844491	A	9/2010
CN	102963222	A	3/2013
CN	103009938	A	4/2013
CN	103171160	A	6/2013
CN	103770579	A	5/2014
CN	103879244	A	6/2014
CN	102015334	B	10/2014
CN	102963222	B	5/2015
CN	102963221	B	6/2015
CN	105365504	A	3/2016
CN	105365505	A	3/2016

CN	103009938	B	6/2016
CN	103879243	B	3/2017
CN	107816564	A	3/2018
CN	108349337	A	7/2018
CN	108349338	A	7/2018
CN	108349340	A	7/2018
CN	109070663	A	12/2018
CN	110114231	A	8/2019
CN	107816564	B	11/2019
CN	108349338	B	6/2020
CN	108349337	B	11/2020
CN	108349340	B	3/2021
CN	109070663	B	8/2021
DE	1042446	B	10/1958
DE	4042446	A1	7/1992
DE	19739144	A1	3/1999
DE	102012106549	A1	5/2014
DE	202017104222	U1	8/2017
DE	102019121377	A1	2/2021
EP	2572903	A1	3/2013
EP	2572903	B1	4/2016
EP	2951036	B1	12/2016
EP	3296131	A1	3/2018
EP	3332223	A1	6/2018
EP	3442813	A1	2/2019
EP	3296131	B1	6/2019
EP	3332223	B1	7/2019
EP	3535142	A1	9/2019
EP	3442813	B1	3/2020
EP	3535142	B1	9/2021
GB	530005	A	12/1940
GB	2089297	A	6/1982
JP	2005231573	A	9/2005
JP	2010149073	A	7/2010
JP	2011203230	A	10/2011
KR	20070098639	A	10/2007
KR	20080039578	A	5/2008
KR	20140003969	U	6/2014
KR	20170029520	A	3/2017
KR	102256754	B1	5/2021
RU	112098	U1	1/2012
WO	2009034321	A1	3/2009
WO	2009110001	A1	9/2009
WO	2012012617	A2	1/2012
WO	2017064106	A1	4/2017

OTHER PUBLICATIONS

"Automatic Trailer Tire Inflation System", Fleet Maintenance, first downloaded Nov. 9, 2023, <https://www.fleetmaintenance.com/in-the-bay/shop-equipment/product/10167762/hendrickson-international-automatic-trailer-tire-inflation-system>.

"Complete tire performance data, seamlessly integrated to your Samsara Fleet Management Solution", Samsara Pressure Pro, first downloaded Oct. 24, 2023, <https://pressurepro.us/samsara/>.

"Continental Designs Tomorrow's Tire", Continental Press Release, Sep. 11, 2019.

"Leveraging Near Field Communication (NFC) to Connect with BLE Smart Sensors", DigiKey, Nov. 16, 2017, <https://www.digikey.com/en/articles/leveraging-nfc-to-connect-with-ble-smart-sensors>.

"The Future of the Tire Management, Meet the New ContiConnect Digital Tire Monitoring", Continental Tire The Americas, LLC, first downloaded Oct. 24, 2024.

"Tiremaxx Tire Pressure Control Systems", Hendrickson, first downloaded Oct. 25, 2023, <https://www.hendrickson-intl.com/products/tiremaxx/tiremaxx>.

Butcher, Lawrence, "Continental launches cloud-based tire monitoring system", Continental TPMS & Electronics, Dec. 9, 2020, <https://www.tiretechnologyinternational.com/news/tpms-electronics/continental-launches-cloud-based-tire-monitoring-system-for-fleet-management.html>.

* cited by examiner

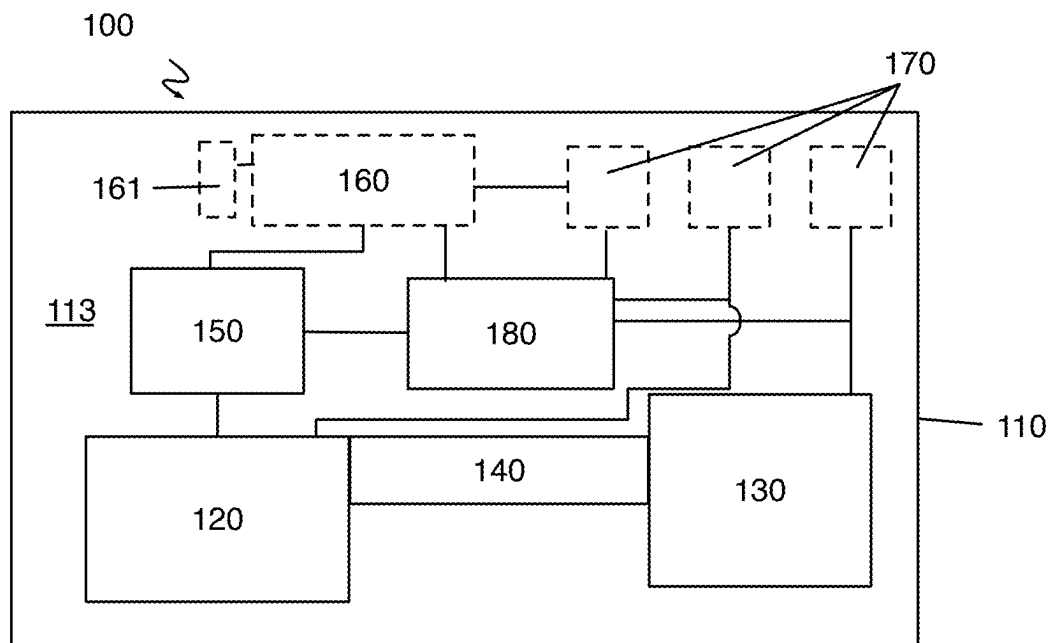


FIGURE 1

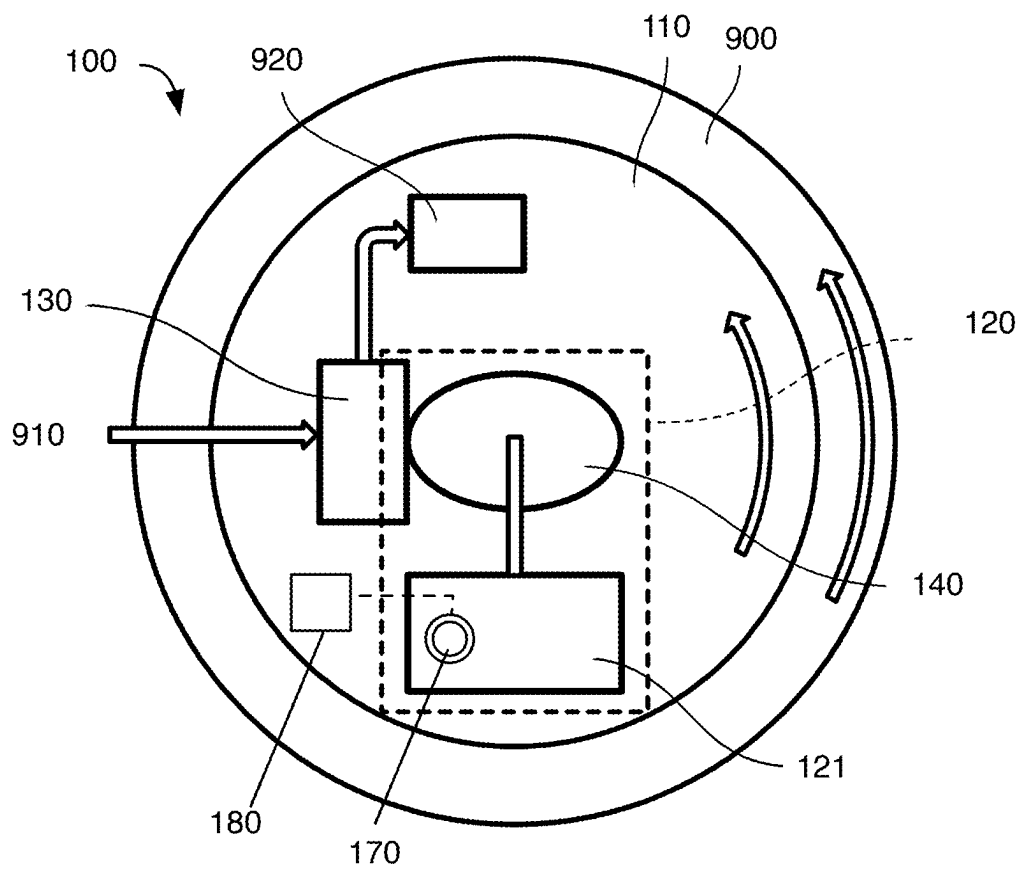


FIGURE 2

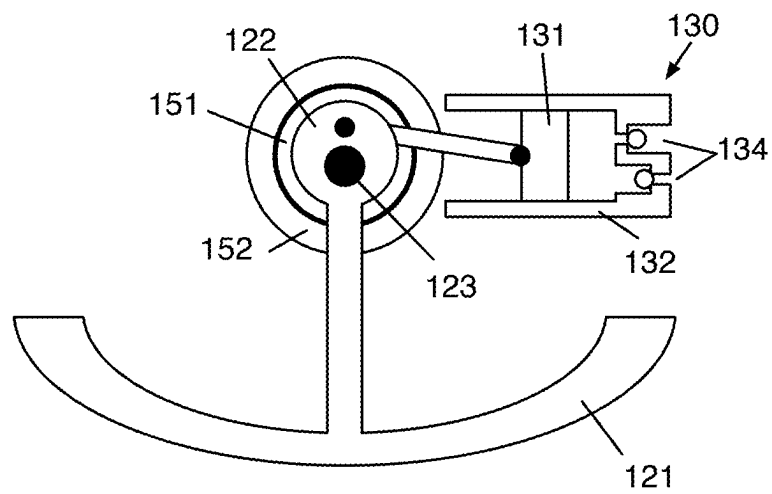


FIGURE 3A

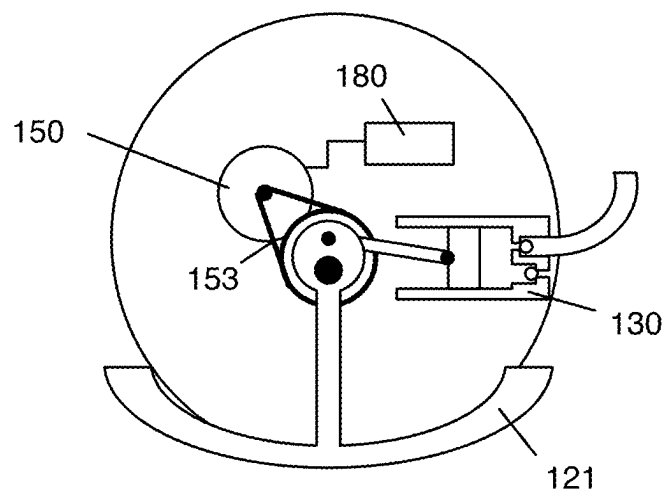


FIGURE 3B

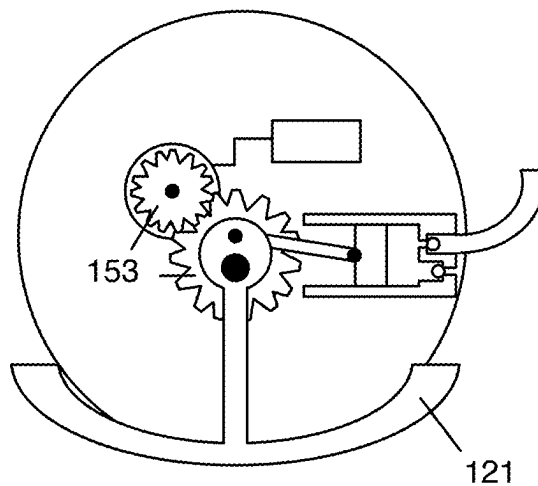


FIGURE 3C

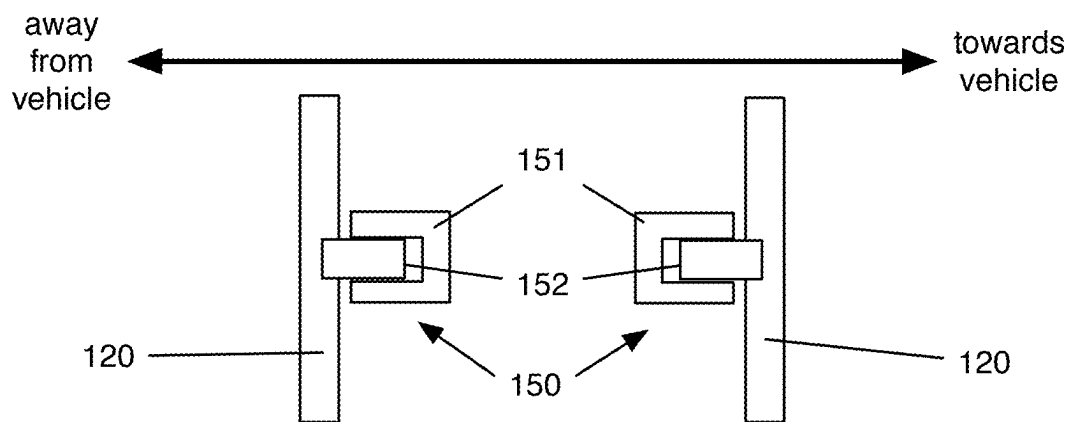


FIGURE 4A

FIGURE 4B

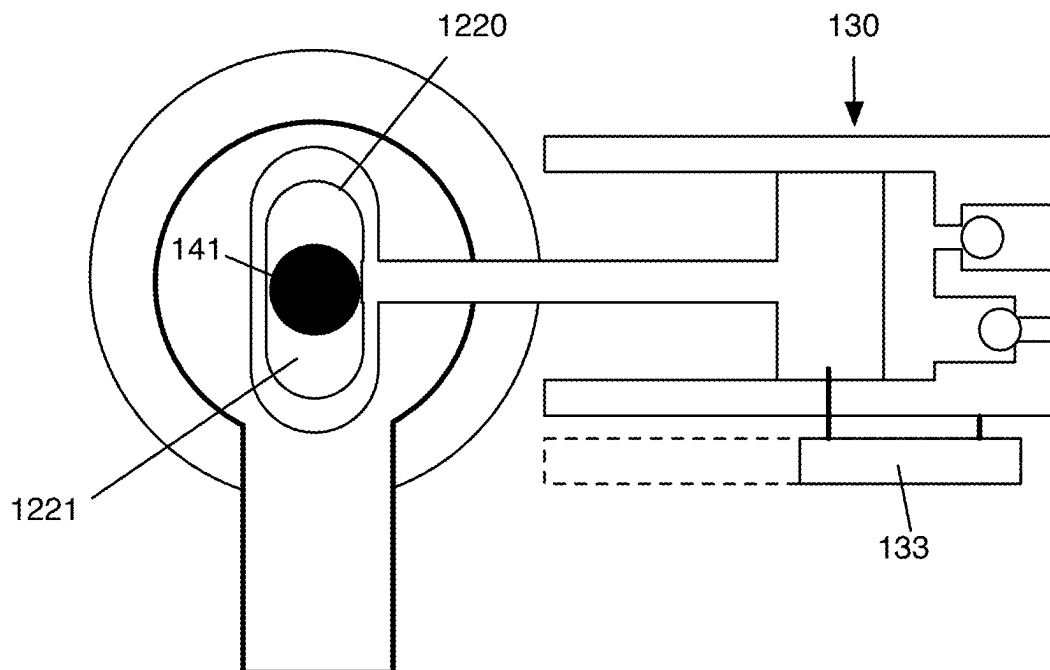


FIGURE 5

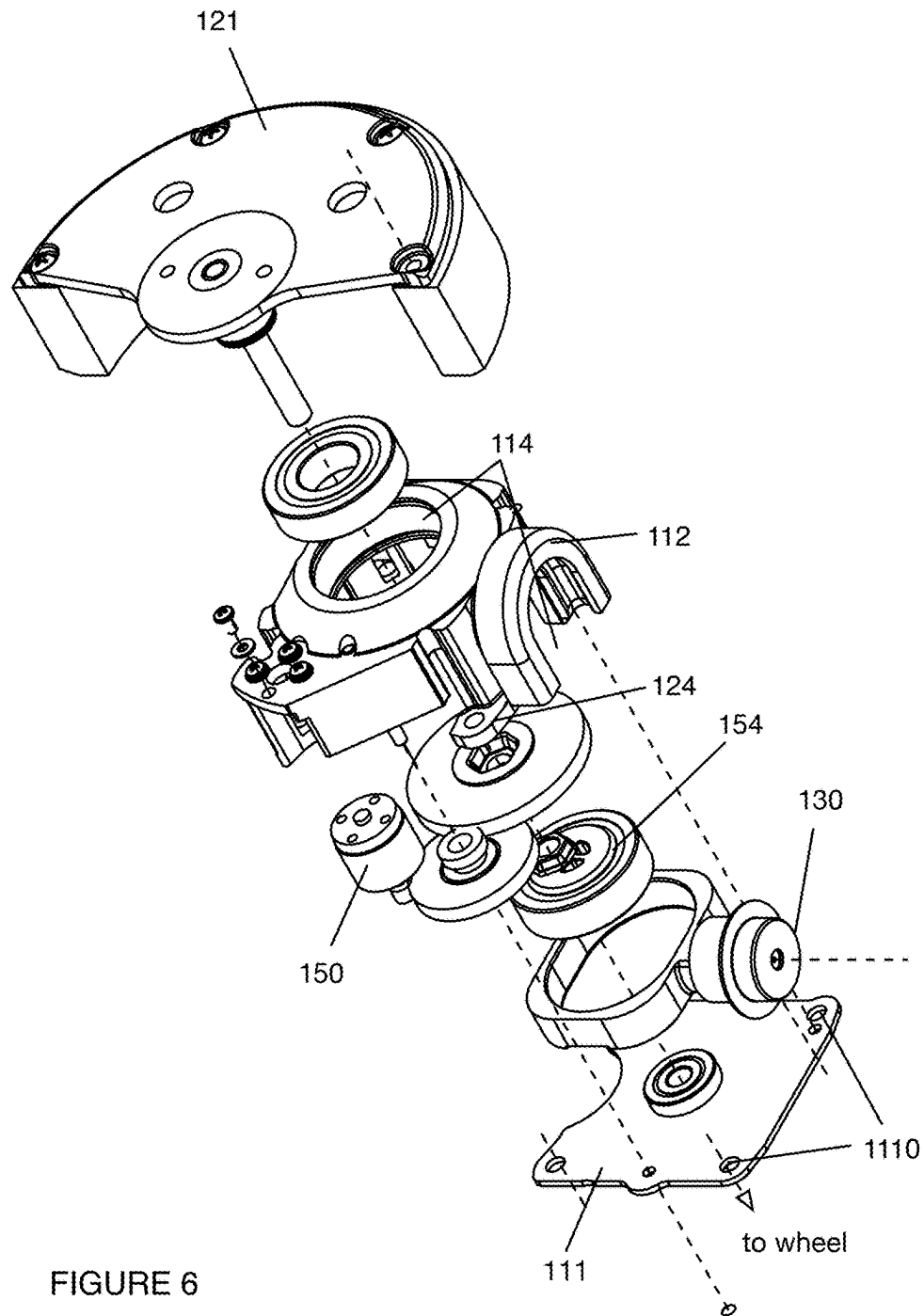
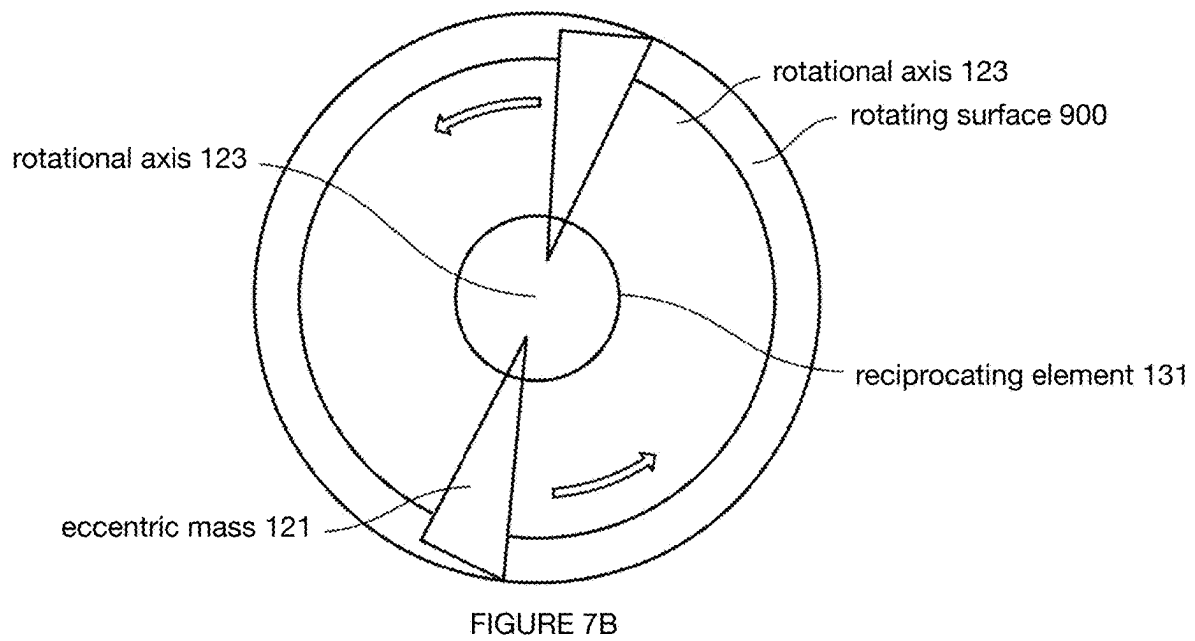
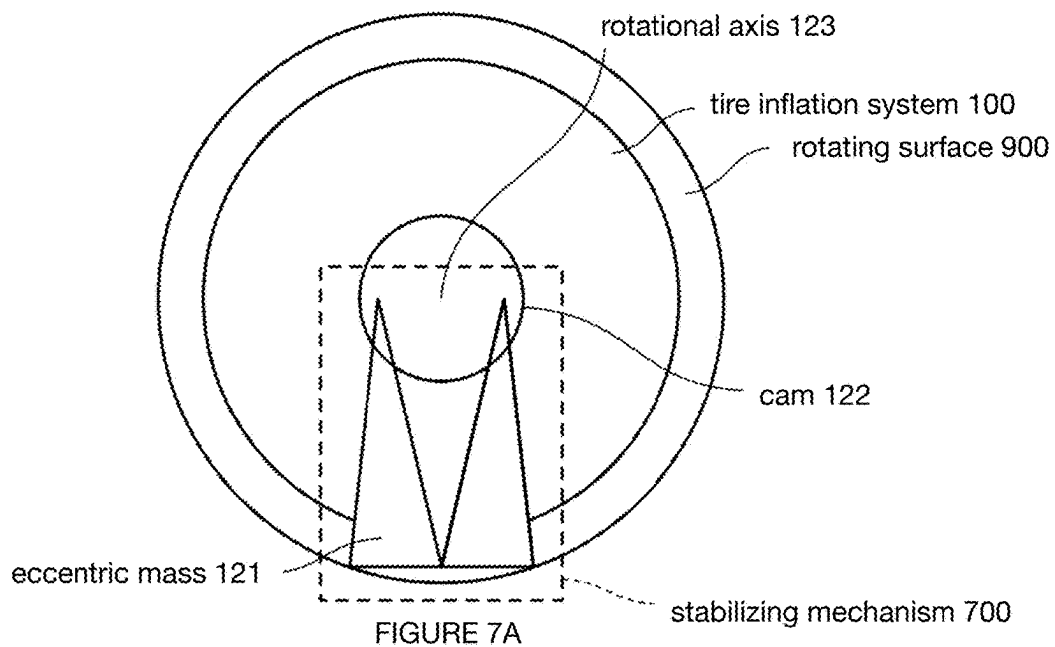
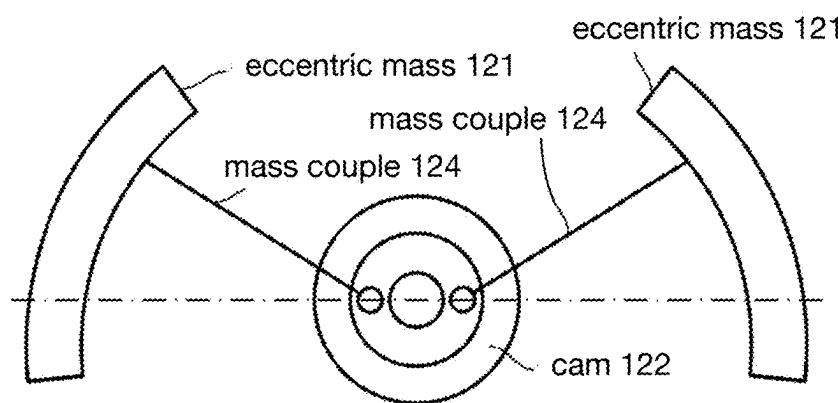
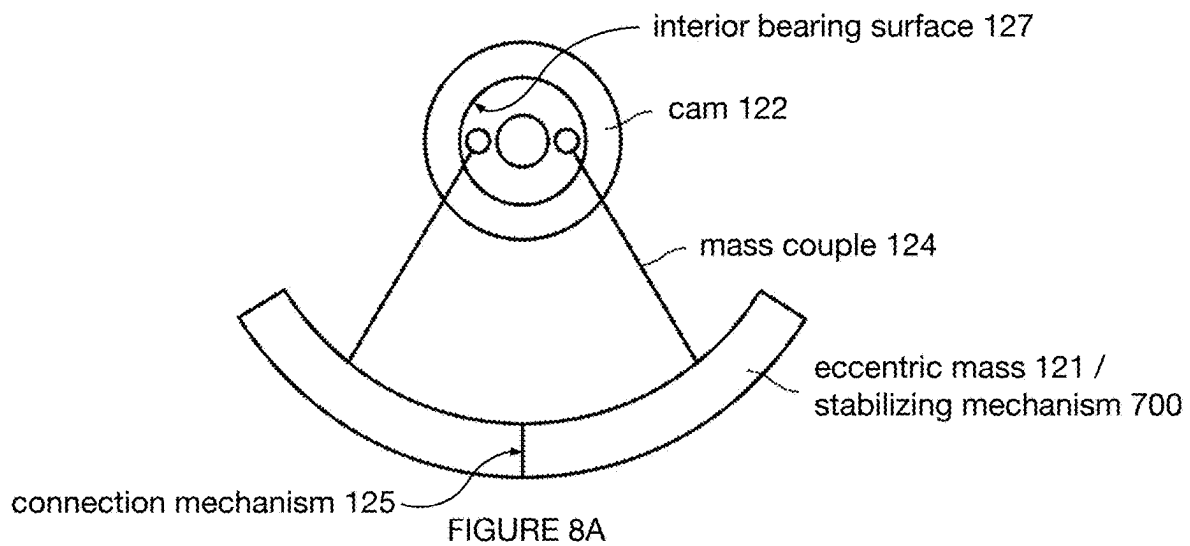


FIGURE 6





SYSTEM FOR TIRE INFLATION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/097,479 filed 16 Jan. 2023, which is a continuation of U.S. patent application Ser. No. 17/868,311, filed 19 Jul. 2022, which is a continuation of U.S. patent application Ser. No. 17/061,313, filed 1 Oct. 2020, which is a continuation-in-part of U.S. patent application Ser. No. 16/276,998 filed 15 Feb. 2019, which is a divisional of U.S. patent application Ser. No. 15/696,816 filed 6 Sep. 2017 which claims the benefit of U.S. Provisional Application Ser. No. 62/383,910, filed 6 Sep. 2016, and U.S. Provisional Application Ser. No. 62/519,061, filed 13 Jun. 2017, each of which is incorporated herein in its entirety by this reference.

U.S. patent application Ser. No. 17/061,313, filed 1 Oct. 2020, is also a continuation-in-part of U.S. patent application Ser. No. 16/161,771 filed 16 Oct. 2018, which is a continuation of U.S. application Ser. No. 15/280,737 filed 29 Sep. 2016, which claims the benefit of U.S. Provisional Application No. 62/235,121 filed 30 Sep. 2015 and is a continuation-in-part of U.S. application Ser. No. 14/839,009 filed 28 Aug. 2015, which is a continuation of U.S. application Ser. No. 14/198,967 filed 6 Mar. 2014, which is a continuation of U.S. application Ser. No. 14/019,941 filed 6 Sep. 2013, which is a continuation of U.S. application Ser. No. 13/797,826 filed 12 Mar. 2013, each of which are incorporated in their entireties by this reference.

TECHNICAL FIELD

This invention relates generally to the pumping field, and more specifically to a new and useful tire-mounted pumping system in the pumping field.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a schematic representation of the tire inflation system.

FIG. 2 is a schematic representation of a variation of the tire inflation system.

FIGS. 3A-3C are schematic representations of example configurations of the tire inflation system.

FIGS. 4A-4B are top views of example relative configurations of the drive mechanism and the torque regulation mechanism.

FIG. 5 is a schematic of an example configuration of a slotted cam configuration of the drive coupler of the tire inflation system.

FIG. 6 is an exploded view of a variation of the tire inflation system.

FIGS. 7A and 7B are schematic representations of a variation of the stabilizing mechanism in the pumping and non-pumping modes, respectively.

FIGS. 8A and 8B are schematic representations of a second variation of the stabilizing mechanism in the pumping and non-pumping modes, respectively.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

The following description of the preferred embodiments of the invention is not intended to limit the invention to these preferred embodiments, but rather to enable any person skilled in the art to make and use this invention.

1. Overview

As shown in FIG. 1, the system for tire inflation 100 includes a drive mechanism 120, a primary pump 130, a drive coupler 140, and a torque regulation mechanism 150.

The system can optionally include an energy storage device 160, one or more sensors 170, and a controller 180. In a variation, the drive mechanism 120 includes a cam 122 and an eccentric mass 121, the primary pump 130 includes a reciprocating element 131 and a pump body 132, and the torque regulation mechanism 150 includes a first portion 151 (e.g., a stator) and a second portion 152 (e.g., a rotor).

The system functions to inflate a tire. The system can also function to translate rotational motion into reciprocating linear motion that can be used to drive a tire inflator (e.g., a pump). The system can also function to translate relative motion between the primary pump 130 and cam 122 into a pumping force, wherein the eccentric mass 121 retains the cam 122 position relative to a gravity vector while the primary pump 130 rotates relative to the cam 122. The system can be operable between several modes, including a pumping (e.g., active) mode and a freewheeling (e.g., passive) mode. In the pumping mode, the tire inflation system 100 preferably pumps an external fluid, such as air, into the tire interior. The external fluid is preferably received from a first reservoir 910 (e.g., the external environment, a canister, etc.) during a recovery stroke of the primary pump 130 and pumped into a second reservoir (e.g., the tire) during a compression stroke of the primary pump 130. However, the fluid can be otherwise suitably pumped. The first reservoir 910 is preferably the ambient atmosphere at a first pressure, and the second reservoir is preferably the tire interior (e.g., bladder) at a second pressure higher than the first pressure. However, the first and second reservoirs can be any other suitable fluid reservoirs at any other suitable absolute and/or relative pressures. In further alternatives, the fluid can be a fluid other than air (e.g., liquid water, pure nitrogen, etc.).

In a first variation of the freewheeling mode, the eccentric mass 121 of the drive mechanism 120 rotates at substantially the same velocity as the wheel (and, thus, as the primary pump 130) such that a negligible (e.g., zero, substantially zero) drive force is supplied by the drive mechanism 120 to the primary pump 130. In the first variation, the torque regulation mechanism 150 can supply a torque to the eccentric mass 121 to excite the mass into rotation about a rotation axis (e.g., of the wheel hub) at the same velocity (e.g., substantially the same, exactly the same) as the wheel, whereupon angular momentum of the eccentric mass 121 substantially maintains the eccentric mass 121 in rotation. Upon excitation of the eccentric mass 121 into rotation at the same velocity as the wheel, the torque regulation mechanism 150 can cease supplying the torque. However, in an alternative implementation of the first variation of the freewheeling mode, the torque regulation mechanism 150 can supply a torque (e.g., continuously, periodically, asynchronously, etc.) to maintain the eccentric mass 121 in rotation about the rotation axis at a suitable velocity such that a negligible drive force is supplied by the drive mechanism 120 to the primary pump 130.

In a second variation of the freewheeling mode, the eccentric mass 121 can be statically connected to the system housing 110 and/or wheel. In the second variation of the freewheeling mode, the eccentric mass 121 rotates along with the housing 110 at the wheel speed, acted upon by a mechanical force supplied by the static connection. For example, the eccentric mass 121 can be clipped, latched, buckled, snapped, or otherwise suitably fastened to the housing 110 and/or any portion of the system or system-

related component rotating along with the wheel (e.g., in a reference frame rotating at the same angular velocity as the wheel). However, the eccentric mass **121** can be otherwise suitably statically connected to the system housing **110** and/or wheel in the second variation of the freewheeling mode.

In a third variation of the freewheeling mode, the eccentric mass **121** can be rotationally decoupled from (e.g., rotates freely relative to) the system housing **110** and/or wheel. In the third variation of the freewheeling mode, rotation of the eccentric mass **121** (e.g., when rotationally decoupled) does not supply a drive force to the primary pump **130** via the cam **122** and drive coupler **140**. For example, the system can include a clutch that can engage and disengage the eccentric mass **121** from the cam **122**, wherein the eccentric mass **121** and the cam **122** are disengaged during system operation in the freewheeling mode, and engaged in the pumping mode. In another example, the system can include a clutch that can engage and disengage the cam **122** from the drive coupler **140**, wherein the cam **122** and the drive coupler **140** are disengaged during system operation in the freewheeling mode, and engaged in the pumping mode. However, in the third variation of the freewheeling mode, the system can additionally or alternatively include any suitable mechanism for rotationally decoupling the drive mechanism **120** from the primary pump **130**.

In a fourth variation of the freewheeling mode, the eccentric mass **121** is maintained at a hanging angle of substantially zero degrees relative to a gravity vector, such that no reciprocating action is produced by the cam **122** upon the reciprocating element **131** of the primary pump **130**. In the fourth variation, the eccentric mass **121** is preferably maintained at a zero hanging angle by the torque regulation mechanism **150**, but can additionally or alternatively be otherwise suitably maintained at a zero hanging angle (e.g., by a locking mechanism).

The system is preferably operable between the pumping and freewheeling modes by the torque regulation mechanism **150** in cooperation with the controller **180**. Controller **180** operation can include generating control instructions based on any suitable control algorithm, and incorporating any suitable sensor inputs. The control instructions can be generated in real-time, near-real time, or at any suitable time. The control instructions and/or parameter values thereof can be selected (e.g., from a database) based on the sensor input values or patterns (e.g., eccentric mass angular kinematics, system lateral kinematics, vehicle kinematics, etc.), calculated (e.g., target operation values calculated based on the sensor input values), optimized (e.g., for pumping, energy harvesting, eccentric mass rotational frequency, etc.), or otherwise determined. However, the system can be otherwise suitably operable between any suitable operating modes by any suitable control and/or regulation mechanism.

The tire inflation system **100** preferably discontinuously inflates the tire (e.g. via periodic inflation, wheel speed-controlled inflation, actively controlled inflation, pressure-dependent inflation, etc.), but may continuously inflate the tire. The tire inflation system **100** is preferably powered by a direct mechanical linkage to the rotating wheel, such that the inflation system pumps fluid into the tire when the tire rotates; but the tire inflation system **100** can alternatively be powered by an actuator that is decoupled from the rotation of the tire (e.g., an electric motor having a separate power source). The tire inflation system **100** can pump fluid using a diaphragm system, a peristaltic system, a piston system, or

any other suitable pumping mechanism. The tire inflation system **100** preferably mounts to a wheel (e.g. to the hub of a wheel), and preferably connects to the tire interior through a valve of the tire. The tire inflation system **100** is preferably configured to be mounted to the wheel for an extended period of time (e.g., on the order of weeks, months, or years); accordingly, removal of the tire inflation system **100** for routine tire pressure checks can be omitted.

2. Benefits

Variants of the systems and/or methods can confer several benefits and/or advantages. First, variants of the system can provide improved resistance to entering an undesired spin condition in which the eccentric mass **121** rotates at substantially the same angular velocity as the wheel (e.g., the freewheeling mode) when reciprocal pumping is desired, by providing torque input to the eccentric mass **121** via the torque regulation mechanism **150**. The torque input can be modulated to provide a counter-force to torque ripple caused by the reciprocating pump (e.g., a back torque acting upon the eccentric mass **121**), to provide a counter-force to large back torques produced by the primary pump **130** during the compression stroke when the system is operated at low vehicle speeds and/or starting from a stopped state (e.g., zero velocity), and/or to provide a counter-force to transient forces resulting from road and/or driving irregularities (e.g., bumps, undulations, vehicle acceleration and deceleration, etc.). This can, in turn, actively increase the amount of time during driving in which the system can usefully pump air using energy harvested from the eccentric mass **121**.

Second variants of the system can enable the tire inflation system **100** to be controllable (e.g., actively controllable) between the pumping and freewheeling modes, by transitioning the pendulum (e.g., eccentric mass **121**) into the freewheeling mode (e.g., wherein the pendulum is rotating at the wheel rotation speed) during periods in which the tire does not require inflation. By operating in the freewheeling mode during periods in which the tire(s) do not require inflation, wear on system components (e.g., reciprocating pump components) can be reduced and the maintenance-free system lifetime can be thus increased. The torque regulation mechanism **150** (e.g., in cooperation with a control system) can also actively transition the system into the pumping mode, by providing a torque against the eccentric mass **121** to control the hanging angle of the eccentric mass **121** relative to a gravity vector (e.g., to stop the eccentric mass **121** from rotating at the wheel rotation speed). In one variation, this includes: determining the wheel rotation speed and controlling electric motor rotor rotation to substantially match the wheel rotation speed. In a second variation, this includes: determining the eccentric mass **121** angle relative to the gravity vector, determining a desired angle, and controlling the electric motor (e.g., the electric motor rotation speed, the angular position of the electric motor, the current or voltage supplied to the electric motor, etc.) to adjust the eccentric mass **121** angle to substantially match the desired angle. However, the system can be otherwise transitioned into the pumping mode. By transitioning into the pumping mode without relying on a passive exit from the freewheeling mode (e.g., due to normal perturbations to the rotary motion arising from road surface irregularities and/or driving behavior), fluid can be provided to the tires on demand (e.g., when tires require immediate inflation, imminent inflation, etc.).

Third, variants of the system can confer several benefits related to on-demand, real-time tire inflation. Tires that are properly inflated improve vehicle fuel economy, and have longer lifetimes before replacement becomes necessary.

Adjustable tire pressure in real or near-real time also enables adjustment of tire parameters (e.g., compressibility) to road and/or environmental conditions. For example, the tire pressure can be increased to take advantage of reduced rolling resistance on recently paved, smooth roads where the risk of a tire rupture due to road roughness is low. In another example, tire pressure can be automatically adjusted to account for changes in ambient pressure and/or temperature, such that an optimal pressure difference between the interior and exterior of the vehicle tire(s) is maintained.

Fourth, variants of the system can be distributed at each wheel of the vehicle (or a subset of wheels of the vehicle), which can reduce the cost of an auto-inflation system compared to a centralized inflation system and can enable the control of tire pressure on a per-wheel basis without the need for complex and expensive plumbing, valve networks, and/or pressurized fluid manifolds. Performing pressurization at the wheel-end can also reduce the likelihood of pressure system failure due to a reduction in the number of pressurized system components, which can be vulnerable to shock and vibration damage when routed beneath a vehicle.

Fifth, variants of the system can be physically rugged, robust, and/or otherwise resilient to the harsh environment in the vicinity of the wheel due to exposure to road debris and other hazards. The placement of variants of the system at the wheel hub area provides a well-suited area for physically shielding system components between the wheel hub and an outer surface of the system housing 110.

However, the system and/or method can confer any other suitable benefits and/or advantages.

3. System

As shown in FIG. 2, the tire inflation system 100 can include: a housing 110; a drive mechanism 120 that includes a cam 122 and an eccentric mass 121; a primary pump 130 that includes a reciprocating element 131, a pump body 132, a return mechanism 133, and one or more inlets 134; a drive coupler 140 that connects the drive mechanism 120 and the primary pump 130; a torque regulation mechanism 150 that includes a rotor and a stator; an energy storage device 160 that includes an energy dissipation mechanism 161; one or more sensors 170; and a controller 180. Variants of the system or components thereof can be similar to the system and/or components described in U.S. application Ser. No. 15/280,737, filed 29 Sep. 2016, incorporated herein in its entirety by this reference.

3.1 Housing

The housing 110 functions to couple system components to a rotating surface 900 (e.g., the hub of a wheel of a vehicle). The housing 110 can also function to mechanically protect (e.g., shield) system components from road debris and other objects that can transiently impact the wheel during vehicle operation. The housing 110 can also function as a mounting substrate for visual indicators of system performance (e.g., for an LED that reports the system status). The housing 110 is preferably removably coupled to a rotating surface 900, such as by way of removable fasteners (e.g., nuts and bolts, screws, brackets, etc.); additionally or alternatively, the housing 110 can be permanently coupled to the rotating surface 900 (e.g., via welding, rivets, permanent fasteners, etc.). The housing 110 is preferably coupled to a rotating surface 900 of the vehicle (e.g., that rotates during vehicle locomotion), and is more preferably coupled to the hub of a vehicle wheel. However, the housing 110 can additionally or alternatively be statically coupled to the rim of a vehicle wheel, a hubcap, to an axle of the vehicle, or any other suitable rotating or non-rotating surface 900 of the vehicle. The housing 110 is preferably coupled to

the vehicle by way of a set of fasteners (e.g., arranged to mate with an existing bolt pattern of the wheel hub), but can additionally or alternatively be integrated directly into the vehicle (e.g., manufactured as part of the wheel hub or vehicle axle) or otherwise suitably attached to the vehicle by any other suitable mechanism. In a specific example, the housing 110 is contiguous with a hubcap of the wheel, and is fastened to the wheel (e.g., via a set of lugnuts) and functions both as a hubcap and the system housing 110. The housing 110 is preferably rotatably coupled to the drive mechanism 120 (e.g., such that the eccentric mass 121 can rotate relative to the housing 110) and statically coupled to the pump body 132 of the primary pump 130 (e.g., such that the primary pump 130 rotates with the housing 110 as the wheel rotates. Alternatively, the housing 110 can be statically coupled to the drive mechanism 120 and rotatably coupled to the primary pump 130, or have any other suitable coupling to the other system components. The housing 110 is preferably substantially rigid, but can additionally or alternatively be flexible, resilient, or have any other suitable structural characteristics. The housing 110 is preferably substantially impermeable to fluids (e.g., waterproof) and can preferably at least partially shield system components from exposure to external liquids (e.g., water splashed onto the wheel from the road surface), but can additionally or alternatively be permeable.

In a first specific example, as shown in FIG. 6, the housing 110 includes an inner housing that includes a first portion 111 (e.g., mounting plate) that defines a hole pattern 1110 arranged to mate with an existing bolt pattern of the wheel, and a second portion 112 that mates with the first portion 111 to cooperatively define a housing lumen 113. The housing lumen 113 contains the primary pump 130, and defines an orifice 114 through which the pump can be connected to a first reservoir 910 of fluid (e.g., ambient air) and a second reservoir of fluid (e.g., the interior of a tire). The housing lumen 113 further contains the torque regulation mechanism 150, which is disposed adjacent to a portion of the drive mechanism 120 such that a torque can be applied by the torque regulation mechanism 150 to the drive mechanism 120 and thereby adjust the angular position of the eccentric mass 121 of the drive mechanism 120 (e.g., to transition the system into the pumping mode or freewheeling mode). In this first specific example, the housing 110 further contains the cam 122, and the eccentric mass 121 is arranged external to the inner housing and coupled to the cam 122 by an axle that extends through the second portion 112. The eccentric mass 121 in this example extends radially past a furthest radial extent of the inner housing, and defines a portion along an arcuate section of the rotational path of the eccentric mass 121 that extends axially toward the first portion in of the inner housing. The housing 110 in this example can further include an outer housing that encloses the eccentric mass and the inner housing.

In a second specific example, the housing 110 is integrated directly with a hubcap of a vehicle wheel, and defines a housing lumen 113. The housing lumen 113 contains the primary pump 130, the drive mechanism 120, the torque regulation mechanism 150, and the drive coupler 140, and is substantially sealed against the external environment. The housing 110 defines an inlet, which can include a shielded cover (e.g., to prevent foreign matter besides air from entrance), through which the primary pump 130 draws ambient air for compression and pumping during system operation. In this second specific example, the eccentric mass 121 is arranged internal to the housing 110. The eccentric mass 121 in this example extends radially toward

an inner surface of the housing lumen **113**, and defines a portion along an arcuate section of the rotational path of the eccentric mass **121**.

3.2 Drive Mechanism

The drive mechanism **120** of the tire inflation system **100** functions to generate a pumping force to drive the primary pump **130**. The drive mechanism **120** can also function to control the magnitude of the pumping force. The drive mechanism **120** preferably includes an eccentric mass **121** and a cam **122**, but can alternatively include any other suitable components for generating the pumping force (e.g., a rotary pump, a diaphragm pump, a turbopump, etc.). The pumping force generated by the drive mechanism **120** is preferably applied in a radial direction relative to the rotational axis **123** of the drive mechanism **120** (e.g., the rotational axis **123** of the wheel), but can alternatively be applied in any suitable direction. The pumping force is preferably applied cyclically (e.g., in a reciprocal manner to drive a reciprocating pump), but can additionally or alternatively be a constant force, a steadily increasing or decreasing force, or have any other suitable temporal profile.

The drive mechanism **120** can be rotatably coupled to the housing **110**, such that the drive mechanism **120** is substantially stationary in a translating reference frame (e.g., translating with the vehicle) as the housing **110** and wheel rotate. The drive mechanism **120** preferably defines a rotational axis **123** about which a portion of the drive mechanism **120** can rotate, and more preferably the cam **122** of the drive mechanism **120** rotates about the rotational axis **123**. However, the rotational axis **123** can additionally or alternatively include the rotational axis **123** about which the eccentric mass **121** rotates, and/or any other suitable axis. The rotational axis **123** of the drive mechanism **120** is preferably coaxial with a rotational axis **123** of the tire inflation system **100** as a whole (e.g., the wheel rotational axis **123**), but can alternatively be offset (e.g., radially offset). The drive mechanism **120** preferably defines a single rotational axis **123** (e.g., about which the cam **122** and eccentric mass **121** rotate), but can alternatively define multiple rotational axes (e.g., a first rotational axis **123** about which the eccentric mass **121** rotates, and a second rotational axis **123** distinct from the first rotational axis **123** about which the cam **122** rotates).

The cam **122** of the drive mechanism **120** functions to mechanically control the magnitude of the pumping force. The cam **122** can also function to convert a torque received from the drive mechanism **120** to a linear force, and apply the linear force against the reciprocating element **131** of the primary pump **130** during the compression stroke. The torque received and/or the linear force applied can be, in variations, constant in time, variable in time, adjustable, or have any other suitable characteristics. In a first variation, the torque provided is modulated in response to a back torque from the reciprocating pump (e.g., assisted by the torque regulation mechanism **150**, defined by a feature of the cam **122**, etc.). The cam **122** preferably defines a bearing surface **1220**, which can be an interior surface of the cam **122**, an exterior surface of the cam **122**, or any suitable combination of interior and exterior surfaces. The bearing surface **1220** can be continuous or discontinuous. In a specific example, as shown in FIG. 5, the bearing surface **1220** is defined within an interior of the cam **122** and includes a slotted lumen **1221**. However, the system can include any suitable cam **122** with any suitable configuration.

The bearing surface **1220** can include a profile that, in variations, defines an arcuate surface, a surface having a

non-uniform curvature, a uniform curvature, and/or any other suitable spatial profile. The profile of the bearing surface **1220** preferably controls the magnitude of the pumping force throughout the compression stroke (e.g., a modulated pumping force, a constant pumping force, etc.). The bearing surface **1220** is preferably arcuate, and preferably has a non-uniform curvature (e.g., an oblong profile or a reniform profile). Alternatively, the bearing surface **1220** can have a uniform curvature (e.g., a circular profile), an angular profile, or any other suitable profile. The bearing surface **1220** preferably includes a compression portion and a recovery portion, corresponding to the compression stroke and the recovery stroke of the primary pump **130**, respectively. The compression portion is preferably continuous with the recovery section, but can alternatively be discontinuous. The bearing surface **1220** preferably has a first section having a high curvature (preferably positive curvature or convex but alternatively negative curvature or concave) adjacent a second section having low curvature (e.g., substantially flat or having negative curvature compared to the first section). The bearing surface **1220** preferably additionally includes a third section connecting the first and second sections, wherein the third section preferably provides a substantially smooth transition between the first and second sections by having a low curvature adjacent the first section and a high curvature adjacent the second section. The compression portion preferably begins at the end of the second section distal the first section, extends along the third section, and ends at the apex of the first section. The compression portion is preferably convex (e.g., when the bearing surface **1220** is an external bearing surface **1220**), but can alternatively be concave. The apex of the first section preferably corresponds to the top of the compression stroke (compressed position). The recovery portion preferably begins at the apex of the first section, extends along the second section, and ends at the end of the second section distal the first section. The recovery portion is preferably substantially flat or concave (e.g., when the bearing surface **1220** is an external bearing surface **1220**), but can alternatively be convex. The end of the second section preferably corresponds to the bottom of the recovery stroke (recovered position). The slope of the compression portion is preferably less than 30 degrees, but can alternatively have any suitable angle. When a roller is used as the force translator, the curvature of the bearing surface **1220** is preferably at least three times larger than the roller curvature or roller diameter, but can alternatively be larger or smaller. However, the bearing surface **1220** can have any suitable profile. The cam **122** is preferably substantially planar with the bearing surface **1220** defined along the side of the cam **122**, in a plane normal to the rotational axis **123** of the cam **122** (e.g., normal the broad face of the cam **122**). The bearing surface **1220** is preferably defined along the entirety of the cam **122** side, but can alternatively be defined along a portion of the cam **122** side. The generated pump force is preferably directed radially outward of the rotational axis **123**, more preferably along a plane normal to the rotational axis **123**. Alternatively, the cam **122** can have a rounded or otherwise profiled edge segment (transition between the cam **122** broad face and the cam **122** side), wherein the bearing surface **1220** can include the profiled edge. Alternatively, the arcuate surface is defined by a face of the cam **122** parallel to the rotational axis **123** of the cam **122**, wherein the generated pump force can be directed at any suitable angle relative to the rotational axis **123**, varying from parallel to the rotational axis **123** to normal to the rotational axis **123**. The compression portion preferably encompasses the majority of the cam **122** profile, but can alternatively encompass

half the cam 122 profile or a small portion of the cam 122 profile. In one variation, the compression portion covers 315 degrees of the cam 122 profile, while the recovery portion covers 45 degrees of the cam 122 profile. However, the compression and recovery portions can cover any other suitable proportion of the cam 122 profile.

The eccentric mass 121 (e.g., pendulum, offset mass) of the drive mechanism 120 functions to offset the center of mass of the drive mechanism 120 from the rotational axis 123 of the drive mechanism 120. The offset functions to retain an angular position of the drive mechanism 120 relative to a gravity vector, in order to generate relative angular motion between the drive mechanism 120 and components statically coupled to the rotating surface 900 (e.g., the housing 110, the pump body 132, etc.). The eccentric mass 121 is preferably a homogenous (e.g., continuous) mass, but can additionally or alternatively be a heterogeneous (e.g., segmented, discontinuous, etc.) mass. In a specific example, as shown in FIG. 3B, the eccentric mass 121 is rotatably attached to the housing 110 at the rotation axis of the wheel and is distributed along a portion of an arc centered at the rotational axis 123. The eccentric mass 121 is preferably a substantially singular, contiguous piece, but can alternatively be made up of multiple pieces and/or segments. In the latter case, the multiple pieces and/or segments are preferably substantially similar in shape, angular and radial position, and mass, but can alternatively be different in profile, mass, angular position, and/or radial position. The eccentric mass 121 can define a curved shape, flat surface, angular shape, and/or any other suitable geometry. At least a portion of the eccentric mass 121 preferably traces an arcuate section of the system perimeter (e.g., aligned with the hub perimeter, inset from the hub perimeter, outside the housing 110 perimeter, inside the housing 110 perimeter, etc.) such that a substantial fraction (e.g., between 10-90%, between 0-100%) of the mass is distributed along the arcuate section. The arcuate section can include any suitable arc (e.g., 90°, 180°, etc.). However, in alternative variations, the eccentric mass 121 can be a spatially confined mass at an end of a pendulum that approximates a point mass. In some variants, the azimuthal distribution of the mass can be varied. For example, the eccentric mass 121 can include articulated arms that can be unfolded outward (e.g., automatically unfolded, manually unfolded, etc.) to distribute the mass along an arcuate section in the azimuthal direction about the rotational axis 123. However, the eccentric mass 121 can be otherwise suitably configured and/or arranged.

The eccentric mass 121 is preferably curved, but can alternatively be substantially flat, angled, or have other suitable shape. The radius of the eccentric mass 121 curvature is preferably maximized, such that the eccentric mass 121 traces an arcuate section of the pump system perimeter. However, the eccentric mass 121 can have any other suitable curvature. The eccentric mass 121 preferably extends at least 90 degrees about the rotational axis 123 of the drive mechanism 120, more preferably 180 degrees about the rotational axis 123, but can extend more or less than 180 degrees about the rotational axis 123. The eccentric mass 121 preferably has substantially more mass than the cam 122, but can alternatively have a substantially similar mass or a smaller mass. The eccentric mass 121 preferably imparts 2 in-lb (0.225 Nm) of torque on the cam 122, but can alternatively impart more or less torque.

The eccentric mass 121 is preferably a separate piece from the cam 122, and is preferably coupled to the cam 122 by a mass coupler 124. Alternatively, the eccentric mass 121

can be incorporated into the cam 122, wherein the eccentric mass 121 is incorporated along the perimeter of the cam 122, incorporated into a half of the cam 122, or incorporated along any other suitable portion of the cam 122. The eccentric mass 121 can be statically coupled to the cam 122 or rotatably coupled to the cam 122. In the variation wherein the eccentric mass 121 is statically coupled to the cam 122, the eccentric mass 121 can be coupled to the cam 122 at the rotational axis 123 of the cam 122, at the rotational axis 123 of the drive mechanism 120, offset from the rotational axis 123 of the cam 122, or at any other suitable portion of the cam 122. The eccentric mass 121 can be permanently connected to the cam 122. Alternatively, the eccentric mass 121 can be transiently connected (removably coupled) to the cam 122, wherein the eccentric mass 121 can be operable between a pumping mode wherein the eccentric mass 121 is coupled to the cam 122 and a non-pumping mode wherein the eccentric mass 121 is disconnected from the cam 122. The mass coupler 124 preferably has a high moment of inertia, but can alternatively have a low moment of inertia. The mass coupler 124 is preferably a disk, but can alternatively be a lever arm, plate, axle, or any other suitable connection. The mass coupler 124 preferably couples to the broad face of the cam 122, but can alternatively couple to the edge of the cam 122, along the exterior bearing surface 1220 of the cam 122, to the interior bearing surface 1220 of the cam 122, to an axle extending from the cam 122 (wherein the cam 122 can be statically fixed to or rotatably mounted to the axle), or to any other suitable portion of the cam 122. The mass coupler 124 can couple to the cam 122 by friction, by a transient coupling mechanism (e.g., complimentary electric or permanent magnets located on the cam 122 and mass coupler 124, a piston, a pin and groove mechanism, etc.), by bearings, or by any other suitable coupling means. When the mass coupler 124 couples to the cam 122 by a transient coupling mechanism, the mass coupler 124 is preferably operable between a coupled mode, wherein the mass coupler 124 connects the eccentric mass 121 to the cam 122, and a decoupled mode, wherein the mass coupler 124 disconnects the eccentric mass 121 from the cam 122. The mass coupler 124 can additionally function as a shutoff mechanism, wherein the mass coupler 124 is switched from the coupled mode to the decoupled mode in response to the detection of a shutoff event (e.g., the reservoir pressure reaching a threshold pressure). In one variation, the mass coupler 124 is a disk located within the lumen defined by an interior bearing surface 1220 of the cam 122, wherein the disk can rotate relative to the interior bearing surface 1220 in the decoupled mode and is coupled to the interior bearing surface 1220 by a friction element in the coupled mode (e.g., the mass coupler 124 acts as a clutch). In another variation, the mass coupler 124 is rotatably mounted on an axle extending from the cam 122 by bearings, wherein the mass coupler 124 can be statically coupled to the cam 122 by one or more sets of magnets or pistons extending from the adjacent broad faces of the cam 122 and mass coupler 124.

3.3 Primary Pump

The primary pump 130 of the tire inflation system 100 functions to pressurize fluid with the pumping force generated by the drive mechanism 120. The primary pump 130 preferably includes a reciprocating element 131 and a pump body 132, and can optionally include a return mechanism 133 and one or more inlets 134. However, the primary pump 130 can include any other suitable components. In variations, the primary pump 130 can function to pressurize the fluid by receiving a reciprocating linear force at the reciprocating element 131. The primary pump 130 is preferably

11

statically mounted to the housing 110, wherein the housing 110 is statically coupled to a rotating surface 900 of the vehicle (e.g., the hub of a wheel). However, the primary pump 130 can additionally or alternatively be statically coupled to a surface that rotates relative to the rotating surface 900 (e.g., that is stationary in an external translating reference frame), such that relative motion is generated between the reciprocating element 131 of the primary pump 130 and the rotating surface 900. The primary pump 130 is preferably positioned radially distal the rotational axis of the drive mechanism 120, but can additionally or alternatively be positioned at least partially coaxially with the rotational axis of the drive mechanism 120 or otherwise suitably arranged. The position of the primary pump 130 relative to the drive mechanism 120 can be fixed or adjustable (e.g., manually adjustable, automatically adjustable, etc.).

In a first variation, the primary pump 130 includes a positive displacement pump wherein the reciprocating element 131 is a piston, and defines a pump cavity (e.g., pump lumen, cylinder) within the pump body 132. In a specific example of this variation, the primary pump 130 is a reciprocating piston pump. In a second variation, the primary pump 130 includes a peristaltic pump. However, the primary pump 130 can include any other suitable pumping mechanism.

The reciprocating element 131 of the primary pump 130 functions to translate back and forth in a reciprocating manner within the pump body 132 to compress fluid transferred from the first reservoir 910 to the second reservoir (e.g., to the tire). The reciprocating element 131 can also function to receive the pumping force from the cam 122 and translate within the lumen of the pump, actuating relative to the pump body 132. This actuation preferably creates a variable pressure within the lumen. The reciprocating element 131 is preferably operable between a compressed position and a recovered position. In the compressed position, a portion of the reciprocating element 131 (e.g., the center) is preferably proximal the pump body 132 bottom. In the recovered position, the portion of the reciprocating element 131 is preferably distal the pump body 132 bottom, and is preferably proximal the pump body 132 opening. The reciprocating element 131 preferably travels along a compression stroke to transition from the recovered position to the compressed position, and travels along a recovery stroke to transition from the compressed position to the recovered position. The reciprocating element 131 can additionally be positioned at a pressurized position, wherein the reciprocating element 131 is located at a second position distal the pump body 132 bottom, wherein the second position is further from the pump body 132 bottom than the recovered position. The reciprocating element 131 is preferably at the pressurized position when the force provided by the lumen pressure exceeds the force provided by the cam 122 on the reciprocating element 131.

The reciprocating element 131 preferably translates along an actuation axis within the primary pump 130 throughout the compression stroke, and can additionally translate along the actuation axis throughout the recovery stroke. The reciprocating element 131 preferably includes an actuating area that provides the pressurization force. The actuating area is preferably the surface area of a broad face of the reciprocating element 131, more preferably the surface area of the broad face proximal the lumen but alternatively any other suitable broad face. Alternatively, the actuating area can be the surface area of a section of the reciprocating element 131 that translates between the compressed position and the recovered position (e.g., the center portion).

12

The reciprocating element 131 preferably forms a fluid impermeable seal with the pump body 132, more preferably with the walls defining the pump body 132 opening, such that the reciprocating element 131 substantially seals the pump body 132 opening. The reciprocating element 131 can be sealed to the pump body 132 by a retention mechanism. The retention mechanism is preferably a clamp that applies a compressive force against the reciprocating element 131 edge and the pump body 132 wall, but can alternatively be screws or bolts through the reciprocating element 131 edge, adhesive between the reciprocating element 131 and the pump body 132 wall or over the reciprocating element 131 and the pump body 132 wall, or any other suitable retention mechanism. The reciprocating element 131 can also be sealed against the pump body 132 wall by melting the interface between the reciprocating element 131 and pump body 132 wall, or by any other suitable means of sealing the reciprocating element 131 against the pump body 132 wall.

The reciprocating element 131 is preferably a flexible diaphragm, but can alternatively be a substantially rigid piston, a piston coupled to the diaphragm, or any other suitable element that actuates in response to the pumping force. The diaphragm is preferably a rolling diaphragm (e.g., with a rolled perimeter, wherein the diaphragm is preferably coupled to the pump body 132 with the extra material distal the lumen) but can also be a flat diaphragm, a domed diaphragm (preferably coupled to the pump body 132 with the apex distal the lumen, but alternatively coupled to the pump body 132 with the apex proximal the lumen), or any other suitable diaphragm.

The pump body 132 functions to cooperatively compress fluid along with the reciprocating element 131. The pump body 132 defines a lumen (e.g., cylinder cavity) in which the fluid is compressed. The pump body 132 is preferably statically mounted to the housing 110, but can be otherwise suitably arranged relative to the housing 110 and/or other system components.

The primary pump 130 can include a return mechanism 133, which functions to bias the reciprocating element 131 in the reverse direction to the direction of the compression stroke during the recovery stroke. The return mechanism 133 preferably provides a recovery force that is less than the compression force provided by the third section of the cam 122, but larger than the force applied by the cam 122 in the second section. The recovery force is preferably provided in a direction substantially parallel to a radial vector extending from the rotational axis of the drive mechanism 120, but can alternatively be provided in any suitable direction. The return mechanism 133 is preferably located on the pump body 132 side of the reciprocating element 131 (distal the cam 122 across the reciprocating element 131), wherein the return mechanism 133 preferably pushes the reciprocating element 131 from the compressed position, through the recovery stroke, and to the recovered position. Alternatively, the return mechanism 133 can be located on the cam 122 side of the reciprocating element 131 (distal the pump body 132 across the reciprocating element 131), wherein the return mechanism 133 pulls the reciprocating element 131 back to the recovered position from the compressed position. The return mechanism 133 is preferably coupled to the perimeter of the reciprocating element 131 or to a component (e.g., a brace) coupled to the reciprocating element 131 and extending past the pump body 132 walls, but can alternatively be coupled to the body of the reciprocating element 131 (e.g., to the section actuating between the compressed position 222 and the recovered position). The return mechanism 133 is preferably coupled to the reciprocating element 131.

13

cating element 131 external the pump body 132, but can alternatively be coupled to the reciprocating element 131 within the pump body 132 240. The return mechanism 133 is preferably a spring, but can also include the intrinsic properties of the actuation element (e.g., the elasticity of the diaphragm) or any other suitable return mechanism 133.

The return mechanism 133 can, in further variations, include an internal spring, an exterior spring (e.g., mounted to an outer surface of the pump body 132), a secondary cam 122 that drives the reciprocating element 131 in opposition to the cam 122 of the drive mechanism 120, and/or any other suitable mechanism.

The primary pump 130 can include one or more inlets 134, which function to receive fluid from the first reservoir 910 into the lumen of the pump body 132 for compression. The inlets 134 can be perpetually open (e.g., fixed orifice 114s in the pump body 132), actuatable (e.g., via controllable valves), shielded (e.g., to protect against influx of foreign matter besides the working fluid), or otherwise suitably constituted.

3.4 Drive Coupler

The drive coupler 140 of the tire inflation system 100 functions to actuate the reciprocating element 131 of the primary pump 130 through the compression stroke as the primary pump 130 rotates about the rotational axis of the wheel. The drive coupler 140 can also function to translate the reciprocating element 131 through the recovery stroke. The drive coupler 140 is preferably coupled between the cam 122 of the drive mechanism 120 and the reciprocating element 131 of the primary pump 130, but can alternatively be otherwise suitably coupled. In a first variation, the drive coupler 140 is coupled to the cam 122 by way of a roller bearing 141 captive within an oblong slot defined by the cam 122, and pinned to the reciprocating element 131 (e.g., rotatable about a fixed point). In a second variation, the drive coupler 140 is pinned to both the cam 122 and the reciprocating element 131. The drive coupler 140 preferably defines an axis having an arcuate position that is fixed relative to the arcuate position the primary pump 130 (e.g., the angular position of the drive coupler 140 about the rotational axis of the wheel is fixed relative to the angular position of the primary pump 130). Preferably, the drive coupler 140 rotates with the primary pump 130 as both components rotate about the rotational axis of the wheel. However, the drive coupler 140 can additionally or alternatively exhibit a different relative rotation to the primary pump 130 (e.g., a different angular velocity, a different trajectory, an off-axis trajectory, etc.).

3.5 Torque Regulation Mechanism

The torque regulation mechanism 150 functions to regulate the torque supplied to the drive mechanism 120 in order to transition the tire inflation system 100 between the pumping and freewheeling operation modes. The torque regulation mechanism 150 can also function to receive torque from the drive mechanism 120 and convert the received torque into electrical potential energy (e.g., to operate as a dynamo). The torque regulation mechanism 150 can also function to provide a torque (e.g., based on instructions from the controller 180) to transition the tire inflation system 100 between the pumping mode and the freewheeling mode, and/or to maintain the tire inflation system 100 in one or more of the pumping mode, the freewheeling mode, and any other suitable operating modes. The torque regulation mechanism is preferably configured to apply a torque based on instructions received from a controller. The instructions can be automatically generated by the controller, generated by a system user in communication with the

14

controller (e.g., manually via an electromechanical interface, wirelessly via a wireless transceiver, etc.), or otherwise suitably generated.

The torque regulation mechanism 150 preferably includes a first portion 151 and second portion 152 that rotate relative to one another, but can be otherwise configured. In one variation, the first portion 151 includes a stator that is statically coupled to a rotating surface 900 (e.g., the housing 110 statically coupled to the wheel) and the second portion 152 includes a rotor that is statically coupled to the eccentric mass 121 such that the rotor rotates along with the eccentric mass 121. In another variation, the stator is statically coupled to the eccentric mass 121 and the rotor is coupled to the rotating surface 900 by way of the housing 110. The rotor and stator are preferably concentrically arranged, but can alternatively be offset (e.g., and mechanically linked by a force transfer mechanism 153). However, the first and second portion 152s of the torque regulation mechanism 150s can be otherwise suitably relatively arranged. In a specific example, the torque regulation mechanism 150 is coupled to the eccentric mass 121 via an intermediate force transfer mechanism 153 (e.g., a gear, a gearbox, a belt, a chain, a clutch, etc.). The torque regulation mechanism 150 is preferably electrically coupled to the controller 180 (e.g., to receive control instructions and/or signals) and the energy storage device 160 by way of one or more direct electrical power and/or data connections. However, the torque regulation mechanism 150 can be otherwise suitably coupled to the controller 180 and/or energy storage device 160.

The torque regulation mechanism 150 is preferably arranged at a different plane from the rotation plane of the eccentric mass 121 (e.g., distal the rotation plane of the eccentric mass 121 in a direction away from the wheel hub, distal the rotation plane of the eccentric mass 121 in a direction toward the wheel hub, etc.). As shown in FIG. 4A, the torque regulation mechanism 150 can be arranged toward the vehicle (e.g., toward the vehicle centerline) relative to the drive mechanism 120 (e.g., the eccentric mass 121 of the drive mechanism 120). As shown in FIG. 4B, the torque regulation mechanism 150 can be arranged away from the vehicle relative to the drive mechanism 120. However, the torque regulation mechanism 150 can additionally or alternatively be arranged in the same plane (e.g., coaxially arranged, offset from the rotation axis of the eccentric mass 121, etc.). In a first variation, as shown in FIG. 3A, the torque regulation mechanism 150 is arranged coaxially with the rotation axis of the wheel and the eccentric mass 121. In further variations, as shown in FIGS. 3B and 3C, the torque regulation mechanism 150 is arranged at an offset position from the rotation axis of the eccentric mass 121, and connected to the eccentric mass 121 via a force transfer mechanism 153 (e.g., a chain and sprocket, a drive belt, etc.). However, the torque regulation mechanism 150 can be otherwise arranged relative to the drive mechanism 120, axis of rotation, or eccentric mass 121. The torque regulation mechanism 153 can apply a: radially inward force, radially outward force, linearly outward force (e.g., away from the wheel or longitudinal vehicle axis), linearly inward force (e.g., toward the vehicle), arcuate force (e.g., within the same plane as eccentric mass rotation), or any other suitable force to the eccentric mass, cam, pump, or other pumping component. The torque regulation mechanism can be statically mounted to: the housing (e.g., interior, exterior, component proximal the tire, component distal the tire, an arcuate segment of the sidewall, etc.), the eccentric mass, the cam, the pump, or to any suitable system component.

15

The torque regulation mechanism **150** preferably includes an electric motor, but can additionally or alternatively include any suitable torque generation and/or regulation mechanism. The electric motor can be an outrunner motor, an inrunner motor, a brushed motor, a brushless motor, an alternating-current motor, a directcurrent motor, a permanent magnet motor, an induction motor, a servo motor, a stepper motor, and/or any other suitable motor. The electric motor preferably generates a rotational force, but can alternatively generate a linear force (e.g., be a linear actuator) or generate any suitable force. In variations, the torque regulation mechanism **150** can include mechanical torque regulation components, such as gears, springs, levers, and any other suitable clockwork components that do not require electrical energy for operation.

The rotor of the torque regulation mechanism **150** functions to move relative to the stator under an applied electromotive force to generate a torque on components statically coupled to the rotor. The rotor can also function to move relative to the stator under an applied torque to generate an electromotive force that can be harvested and stored as electrical potential energy (e.g., at the energy storage device **160**). In a first variation, the rotor is statically coupled to a surface that rotates with the wheel. In a second variation, the rotor is statically coupled to a surface that is substantially stationary relative to the wheel. However, the rotor can be otherwise suitably coupled.

The stator of the torque regulation mechanism **150** functions to move relative to the rotor under an applied electromotive force to generate a torque on components statically coupled to the stator. The stator can also function to move relative to the rotor under an applied torque to generate an electromotive force that can be harvested and stored as electrical potential energy (e.g., at the energy storage device **160**). In a first variation, the stator is statically coupled to a surface that is substantially stationary relative to the wheel. In a second variation, the stator is statically coupled to a surface that rotates with the wheel. However, the stator can be otherwise suitably coupled.

The torque regulation mechanism **150** can include an engagement mechanism **154** that functions to mechanically engage and/or disengage the eccentric mass **121** from other system components. For example, the engagement mechanism **154** can include a clutch that mechanically engages the eccentric mass **121** and the drive coupler **140** during system operation in the pumping mode (e.g., such that a drive force is provided by the eccentric mass **121** when the eccentric mass **121** is maintained at a non-zero hanging angle), and that mechanically disengages the eccentric mass **121** and the drive coupler **140** during system operation in the freewheeling mode (e.g., such that no drive force is provided by the eccentric mass **121** irrespective of the angular position and/or velocity of the eccentric mass **121**). In some variations, the mass coupler **124** can function as an engagement mechanism **154**. However, the engagement mechanism **154** can include any other suitable mechanism for mechanically retaining the eccentric mass **121** relative to the pump and/or other rotating components of the system.

In a first specific example, as shown in FIG. 6, the torque regulation mechanism **150** includes an electric motor wherein the stator of the electric motor is rigidly attached to the eccentric mass **121** (e.g., an arcuate segment of the stator defines a portion of the eccentric mass **121**), the rotor of the electric motor is rigidly coupled to a rotating surface **900** (e.g., the housing **110**, the wheel hub, via mounting components, directly coupled via a weld, etc.), and the rotor is connected to the drive coupler **140** that drives the primary

16

pump **130**. In a second specific example, the torque regulation mechanism **150** includes an electric motor wherein the stator is rigidly mounted to the housing **110**, and is offset from the tire inflation system **100**'s rotational axis and is connected to the eccentric mass **121** by a force linkage (e.g., a gearbox).

3.6 Energy Storage Device

The tire inflation system **100** can include an energy storage device **160**, which functions to provide power to the torque regulation mechanism **150**. The energy storage device **160** can also function to receive power from the torque regulation mechanism **150** (e.g., when the torque regulation mechanism **150** is operating as a dynamo). The energy storage device **160** can, in some variations, function to store compressed fluid generated by the primary pump **130** (e.g., in a compressed air canister). The energy storage device **160** is preferably coupled to the torque regulation mechanism **150** (e.g., via a direct electrical connection for power provision and/or reception), but can additionally or alternatively be coupled to the controller **180**, primary pump **130**, and/or any other system components. The system preferably includes a single energy storage device **160**, but can additionally or alternatively include redundant (e.g., multiple) energy storage device **160**s (e.g., to provide backup power to system components such as the torque regulation mechanism **150**). The energy storage device **160** is preferably coupled to the housing **110** and rotates with the wheel, but can alternatively be coupled to the eccentric mass **121** or to any other suitable system component. The energy storage device **160** is preferably arranged axially inward (e.g., along the direction of the vehicle axle) from the eccentric mass **121**, but can alternatively be arranged axially outward from the eccentric mass **121**. In a first variation, the energy storage device **160** includes a battery. In further variations, the energy storage device **160** can include a super capacitor, a compressed air canister, one or more springs, and/or any other suitable energy storage mechanisms.

The energy storage device **160** can optionally include an energy dissipation mechanism **161** that functions to dissipate excess energy generated by the torque regulation mechanism **150** (e.g., when the torque regulation mechanism **150** is operating as a dynamo) in cases wherein the energy storage device **160** is at full capacity (e.g., when the battery is fully charged). For example, the energy dissipation mechanism **161** can include an electrical resistor, a resistor network, and/or any other suitable passive component for dissipating electrical energy in variations wherein the energy storage device **160** includes an electrical energy storage device **160** (e.g., a battery, capacitor, supercapacitor, etc.). In another example, the energy dissipation mechanism **161** can include an active energy dissipation mechanism **161**, such as a fan, water pump, light emitting element, and/or any other suitable powered mechanism, to utilize excess recovered energy stored at the energy storage device **160** (e.g., for the purpose of cooling, user notification generation, etc.).

3.7 Sensors

The tire inflation system **100** can include one or more sensors **170**, which function to sense operational parameters of the system (e.g., tire pressure, whether the system is in an "on" state or an "off" state, whether the system is operating within nominal ranges, etc.). The sensors **170** can also function to provide sensor data to a controller **180**. The sensors **170** can also function to detect, in cooperation with the controller **180**, whether the system is operating in the freewheeling and/or pumping modes (e.g., by comparing a measured rotational velocity of the eccentric mass **121** with a measured rotational velocity of the wheel). The system can

include one or more pressure sensors **170**, which can be connected to the output of the primary pump **130** to monitor the pressure of the fluid provided to the tire. The sensor(s) are preferably connected to the controller **180** (e.g., via a signal pathway) to provide sensor data (e.g., sensor signals) to the controller **180**, and mounted on and/or within the housing **110** (e.g., for mechanical support). However, the sensor(s) can be otherwise suitably connected. System sensors **170** can include pressure sensors **170** (e.g., capacitively-based diaphragm deflection gauges), flow rate sensors **170**, mass flow sensors **170** (e.g., inline impellers), orientation sensors **170** (e.g., accelerometers, inertial measurement units, gyroscopes, etc.), rotary encoders, and/or any other suitable type of sensor.

In a first variation, the system includes a pressure sensor arranged at an interface between the output of the primary pump **130** and the inlet of the second reservoir (e.g., the tire) to continuously measure the static pressure of the second reservoir. In a second specific example, the system includes a rotary encoder coupled to the torque regulation mechanism **150** that periodically measures (e.g., at 1 kHz) the angular position of the rotor of the torque regulation mechanism **150**.

In a second variation, the system includes a sensor (e.g., an angular position sensor, angular velocity sensor, etc.) communicatively coupled to the controller **180**, that detects and outputs the relative angular velocity between the eccentric mass **121** and the primary pump **130**, wherein the controller **180** generates instructions based on an output of the differential angular velocity sensor and controls the torque regulation mechanism **150** based on the instructions. The sensor can be arranged to be in contact with the eccentric mass **121** (e.g., a contact sensor) and output the angular position and/or velocity based a characteristic of the contact; for example, the electrical resistance through a portion of the sensor can increase or decrease based on the position at which the sensor contacts the eccentric mass **121**. The sensor can additionally or alternatively be physically separated from the eccentric mass **121**; for example, the sensor can include an optical sensor that counts the frequency of optical occlusions of the optical sensor by the eccentric mass **121** during rotation of the eccentric mass **121** and/or the wheel, from which the angular velocity of the eccentric mass **121** can be computed (e.g., by the controller **180**). However, the system can include any other suitable sensors **170**.

3.8 Controller

The tire inflation system **100** can include a controller **180**, which functions to generate control in puts in response to received sensor data and/or instructions. The controller **180** can also function to control the torque regulation mechanism **150** to operate the system between operating modes (e.g., the pumping mode, the freewheeling mode, etc.). The controller **180** can also function to adjust the pressure set point of the tire inflation system **100** and to control the tire inflation system **100** to maintain the tire pressure at the pressure set point. The controller **180** can also function to generate messages in response to system behavior (e.g., error codes). The controller **180** can be communicatively coupled to the sensors **170** and torque regulation mechanism **150** of the system, and in some variations can be communicatively coupled to a remote computing system (e.g., a vehicle ECU, a mobile device within the vehicle, etc.) via a communication system (e.g., wired communications system; wireless communications system, such as Bluetooth, WiFi, Zigbee, cellular, etc.).

In a first specific example of controller **180** operation, the controller **180** detects a perturbative torque to the eccentric

mass **121** (e.g., from an angular position sensor of the eccentric mass **121**) and controls the torque regulation mechanism **150** (e.g., an electric motor) to dampen the perturbative torque and maintain the system in the pumping mode (e.g., maintain a non-zero angle between a gravity vector and the eccentric mass **121**) and prevent the perturbative torque from transitioning the system from the pumping mode into the freewheeling mode. The perturbative torque can be detected via heuristic comparisons (e.g., pattern matching), deterministic comparisons (e.g., an oscillation exceeding a threshold perturbation magnitude), and/or in any other suitable manner. The perturbative torque can be a back torque generated by the primary pump **130** (e.g., due to reciprocating pump dynamics), a disturbance originating from road roughness, external forces and/or shocks, or any other source of torque or force.

In a second specific example of controller **180** operation, the controller **180** receives an instruction (e.g., from a vehicle control system, a user, etc.) to cease pumping (e.g., to transition the system into the freewheeling operating mode from the pumping mode), and in response controls the torque regulation mechanism **150** to apply a torque to the eccentric mass **121** to induce rotation of the eccentric mass **121** at substantially the same angular velocity of the rotating wheel.

3.9 System Examples

In a first specific example of the tire inflation system **100**, the system includes a drive mechanism **120**, a pump, a drive coupler **140**, a torque regulation mechanism **150**, and a controller **180**. The drive mechanism **120** defines a rotational axis, and includes a cam **122** rotatable about the rotational axis and an eccentric mass **121** coupled thereto that offsets a center of mass of the drive mechanism **120** from the rotational axis along the radial vector. The pump is arranged radially distal the rotational axis of the drive mechanism **120**, and includes a chamber defining a chamber lumen, and a reciprocating element **131** arranged at least partially within the chamber lumen and translatable along a pump axis normal to the rotational axis. The drive coupler **140** is coupled between the cam **122** at a first position and the reciprocating element **131** at a second position. The first position is radially distal the rotational axis about which the cam **122** rotates, and the second position is fixed (e.g., pinned) to the reciprocating element **131**. The torque regulation mechanism **150** (e.g., an electric motor, a clockwork mechanism, etc.) includes a first portion rigidly coupled to the eccentric mass **121**, and a second portion rotatably coupled to the first portion. The controller **180** is communicatively coupled to the torque regulation mechanism **150** (e.g., by a hardwire data connection, serial data port, etc.), and to a sensor (e.g., an angular velocity sensor, an angular position sensor, a rotary encoder, etc.) that senses an angular characteristic (e.g., angular position, angular velocity, etc.) of the eccentric mass **121** relative to a gravity vector. The controller **180** is configured to operate the system between the pumping mode and the freewheeling mode. In the pumping mode, the torque regulation mechanism **150** maintains the eccentric mass **121** at a hanging angle (e.g., defined by the radial vector between the rotational axis and an end of the eccentric mass **121** opposing the point at which the eccentric mass **121** is connected to the rotational axis) greater than 0° relative to the gravity vector. In cases where the system is in the freewheeling mode prior to transitioning to the pumping mode, the torque regulation mechanism **150** applies a torque to the eccentric mass **121** to stimulate the eccentric mass **121** to exit a spin condition (e.g., wherein the eccentric mass **121** is rotating about the rotational axis at

substantially the same angular velocity as the wheel to which the system is attached).

In a related specific example, the system further includes an energy storage device **160** (e.g., a battery, a torsional spring, a pneumatic cylinder, etc.) communicatively coupled (e.g., via a direct electrical connection, a direct mechanical connection, a fluid connection, etc.) to the torque regulation mechanism **150** and the controller **180**. The energy storage device **160** is operably between a harvesting mode and a powering mode. In the harvesting mode, the energy storage device **160** receives and stores energy harvested from the torque regulation mechanism **150**, which in turn receives a torque input from the eccentric mass **121**. For example, in a case where the eccentric mass **121** is decoupled from the cam **122** and/or primary pump **130** (e.g., such that zero drive force is provided to the reciprocating element **131**), the eccentric mass **121** can be maintained at a non-zero hanging angle relative to a gravity vector in order to harvest energy (e.g., gravitational energy) to store within the energy storage device **160**. In the powering mode, the energy storage device **160** provides energy to the torque regulation mechanism **150**, which in turn provides a torque input to the eccentric mass **121**. The system and the energy storage device **160** of this example is preferably operated between the harvesting and powering mode by way of the controller **180**, but can be additionally or alternatively operated by any suitable mechanism or control instructions.

The systems and methods of the preferred embodiment and variations thereof can be embodied and/or implemented at least in part as a machine configured to receive a computer-readable medium storing computer-readable instructions. The instructions are preferably executed by computer-executable components preferably integrated with the system and one or more portions of the processor and/or the controller **180** **430**. The computer-readable medium can be stored on any suitable computer-readable media such as RAMs, ROMs, flash memory, EEPROMs, optical devices (CD or DVD), hard drives, floppy drives, or any suitable device. The computer-executable component is preferably a general or application specific processor, but any suitable dedicated hardware or hardware/firmware combination device can alternatively or additionally execute the instructions.

In a second specific example, the tire inflation system **100** includes a drive mechanism **120**, a drive coupler **140**, a reciprocating pump, a torque regulation mechanism **150**, a housing **110**, and a controller **180**. The drive mechanism **120** defines a rotational axis, and includes a cam **122** rotatable about the rotational axis and an eccentric mass **121** coupled to the cam **122** that offsets a center of mass of the drive mechanism **120** from the rotational axis along a radial vector and is rotatable about the rotational axis. The cam **122** includes a slotted lumen that defines an interior surface. The drive coupler **140** defines a first and second end, and the first end of the drive coupler **140** is coupled to the cam **122** at a position radially distal the rotational axis. The first end further includes a roller bearing, and at the coupling position to the cam **122** the roller bearing is captivated within the slotted lumen and contacts the cam **122** at the interior surface. The second end is coupled to the reciprocating pump. The reciprocating pump is arranged radially distal the rotational axis of the drive mechanism **120**, and includes a pump body **132** (e.g., chamber) that defines a chamber lumen, and a reciprocating element **131** arranged at least partially within the chamber lumen and translatable along a pump axis defined longitudinally along the chamber lumen and is normal to the rotational axis (i.e., the pump axis is

perpendicular to the rotational axis of the system). The reciprocating pump further includes a return mechanism **133** (e.g., a spring) that applies a return force (e.g., a spring force) that biases the reciprocating element **131** along the pump axis towards the uncompressed position (e.g., away from the base of the chamber lumen). The return mechanism **133** is fixed to the reciprocating element **131** and the chamber, and is arranged externally to the chamber lumen (e.g., on the outside of the pump body **132**). The torque regulation mechanism **150** applies a controllable torque to the eccentric mass **121**, such that the controllable torque urges rotation of the eccentric mass **121** about the rotational axis. The torque regulation mechanism **150** further includes an engagement mechanism **154** (e.g., a clutch) that engages the eccentric mass **121** in at least a first and second configuration. In the first configuration, the eccentric mass **121** is mechanically coupled to the primary pump **130** (e.g., via the cam **122** and the drive coupler **140**), whereas in the second configuration, the eccentric mass **121** is mechanically decoupled from the primary pump **130** (e.g., via rotational decoupling from the cam **122**, mechanical decoupling from the drive coupler **140**, etc.). The torque regulation mechanism **150** further includes an electric motor and a force transfer mechanism **153** (e.g., a gearbox), and the force transfer mechanism **153** is coupled between the electric motor and the eccentric mass **121** such that torques transmitted between the electric motor and the eccentric mass **121** are mechanically transferred through the gears of the gearbox. The electric motor is arranged at an offset position from the rotational axis, and the rotational axis of the electric motor (e.g., defined by an output shaft of the motor) is parallel to the rotational axis of the eccentric mass **121**. The housing **110** retains the pump, the torque regulation mechanism **150**, and the cam **122**. The eccentric mass **121** is arranged external to the housing **110** and coupled to the cam **122** by a fixed axle that extends through an orifice **114** of the housing **110** along the rotational axis. A portion of the eccentric mass **121** is rotatable about the rotational axis along a circular path, the circular path having a radius greater than the farthest radial extent of the housing **110** (e.g., outside the perimeter of the housing **110**). The eccentric mass **121** is distributed along an arcuate section of the circular path (e.g., a 90° section, a 180° section, etc.).

In the second specific example above, the controller **180** is communicatively coupled to the torque regulation mechanism **150** (e.g., via a direct electrical connection, a serial data connection, a parallel data connection, a wireless data connection, etc.) and generates instructions which are provided (e.g., by the controller **180**) to the torque regulation mechanism **150** to operate the system between a first and second mode. In the first mode, the drive coupler **140** is mechanically engaged with the eccentric mass **121**, such that a drive force is provided to the primary pump **130** by the relative motion between the eccentric mass **121** and the primary pump **130**. In the second mode, the drive coupler **140** is mechanically disengaged from the eccentric mass **121**, such that no drive force is provided to the primary pump **130** by the relative motion (e.g., in cases where relative motion occurs) between the eccentric mass **121** and the primary pump **130** or by any other kinematic mechanism.

Although omitted for conciseness, the preferred embodiments include every combination and permutation of the various system components and/or method blocks.

The system **100** can additionally include a stabilizing mechanism **700** that functions to reduce rotational surface imbalance when the eccentric mass **121** becomes excited (e.g., begins spinning) when the system **100** rotates at or

21

near the excitation frequency of the eccentric mass **121**. The stabilizing mechanism **700** is preferably the eccentric mass **121**, wherein the eccentric mass **121** is collectively formed from multiple sections. However, the stabilizing mechanism **700** can alternatively be any other suitable stabilizing mechanism **700**. When the eccentric mass **121** begins to spin, the composite sections of the eccentric mass separate. This is particularly useful when system oscillations cause the eccentric mass **121** (and positioning mechanism) to spin about the shaft; the centrifugal forces cause the sections of the split eccentric mass **121** to separate and be evenly distributed about the axis of system rotation, as shown in FIGS. 7A and 7B. Not only does this have the effect of dynamically balancing the system and/or rotating surface **900**, but the even distribution of the eccentric mass **121** within the system also halts system pumping. The latter effect can allow the eccentric mass **121** to additionally function as a control mechanism, wherein the eccentric mass resonant frequency can be tailored such that pumping is ceased when a predetermined rotation speed or vibration frequency is reached. The multiple sections are preferably each positioned the same radial distance away from the rotational axis (the eccentric mass **121** is radially divided into multiple sections, wherein the multiple sections have different angular positions), but can alternatively be positioned at different radial distances (e.g., wherein the multiple sections have substantially similar angular positions, etc.). The multiple sections preferably share a common plane, wherein the common plane is preferably substantially parallel to the rotational surface. The multiple sections can collectively form an arc, centered about the rotational axis, that intersects the common plane (e.g., the multiple sections are adjacent along an arc), form a block that intersects the common plane, or collectively form any other suitable structure. Alternatively, the multiple sections can be stacked along the thicknesses of the sections, wherein the section thicknesses are preferably parallel to the rotational axis. The multiple sections preferably have substantially the same mass, but can alternatively have different masses. The center of mass for each eccentric mass section is preferably offset from the mass couple connection point for each eccentric mass section, and is preferably arranged proximal an adjacent eccentric mass section. In operation, the eccentric mass sections separate until the centers of mass of the eccentric mass sections oppose each other across the axis of rotation.

When the eccentric mass **121** is cooperatively formed by multiple sections (e.g., a first eccentric/hanging mass and a second eccentric/hanging mass, etc.), the mass couple **124** (e.g., a mass coupler) preferably also includes multiple sections, wherein each mass couple section statically couples to an eccentric mass section. The mass couple sections are preferably rotatably coupled to the cam **122**, but can alternatively be statically coupled to the cam **122**. Each mass couple section is preferably rotatably coupled to the remaining mass couple sections, but can alternatively be statically coupled to one or more of the remaining mass couple sections. In one variation as shown in FIGS. 8A and 8B, the end of each mass couple section opposing the eccentric mass section is rotatably coupled to the housing. The angular positions of mass couple section ends are preferably static relative to the housing, wherein the mass couple section ends are preferably equally distributed about the axis of rotation. In another variation, the end of each mass couple section opposing the eccentric mass section includes a bearing, wherein the bearing is slidably engaged within a circumferential groove statically coupled to the cam **122** and encircling the rotational axis. When the rotation

22

frequency of the rotating surface **900** is below or above the excitation frequency for the cooperatively defined eccentric mass **121**, the centrifugal force of the rotation preferably retains the eccentric mass sections (and mass couple sections) in substantially adjacent positions. When the rotation frequency of the rotating surface **900** is at the excitation frequency, the centrifugal force preferably causes the bearings to slide within the groove, distributing the multiple eccentric mass sections substantially equally about the rotational axis. The bearings and/or the eccentric mass sections can each additionally include magnets, disposed in repulsive relation to adjacent magnets, which facilitate eccentric mass separation in response to the receipt of a system oscillation. In another variation, the mass couple sections rotatably couple along the longitudinal axis of an axle extending from the cam **122** (e.g., mass couple sections are stacked along the axle). In another variation, one mass couple section is statically connected to the cam **122** while the remaining mass couple sections are rotatably connected to the cam **121**. However, the mass couple sections can be otherwise connected to the cam **122**.

When the mass couple **124** couples to the cam **122** (e.g., a positioning subcomponent) at the rotational axis, the mass couple **124** is preferably operable between the coupled mode, wherein the mass couple **124** connects the eccentric mass **121** to the cam **122**, and the decoupled mode, wherein the mass couple **124** disconnects the eccentric mass **121** from the cam **122**. In one variation, the mass couple **124** is a disk located within the lumen defined by an interior bearing surface **127** (e.g., a cylindrical inner surface) of the cam **122**, wherein the disk can rotate relative to the interior bearing surface in the decoupled mode and is coupled to the interior bearing surface by a friction element in the coupled mode. The mass couple sections are preferably rotatably coupled to the disk, but can alternatively be disk sections (e.g., concentric circles, arcuate pieces, etc.). The friction element can be a high-friction coating along the interior bearing surface, a high-friction coating along the mass couple **124** exterior, a roller or wedge, or any other suitable element capable of providing friction between the interior bearing surface and the mass couple **124**. The friction element is preferably selected such that the cooperative centrifugal force of the eccentric mass **121** in the coupled mode applies sufficient force to the mass couple **124** such that the friction between the mass couple **124** and the interior bearing surface retains the mass couple position relative to the cam **122**. The friction element is preferably selected such that the cooperative centrifugal force of the eccentric mass sections in a separated or decoupled mode does not provide enough force to interface friction to retain the mass couple position relative to the cam **122**, thereby allowing free mass couple rotation. In another variation, the mass couple **124** is rotatably mounted on an axle extending from the cam **122** by bearings, wherein the mass couple **124** can be statically coupled to the cam **122** by one or more sets of magnets or pistons extending from the adjacent broad faces of the cam **122** and mass couple **124**. However, the static mass couple connection to the cam **122** to achieve the coupled mode can be selectively controlled by any other suitable passive or active means.

The eccentric mass **121** can additionally include a connection mechanism **125** (e.g., a coupler) that functions to couple the multiple sections together. The connection mechanism is preferably located on the interfaces of adjacent sections, but can alternatively be located within the section bodies, at the interfaces of adjacent mass couple sections, or at any other suitable location. The coupling force

23

of the connection mechanism is preferably selected such that it is substantially equal to or lower than the angular separation force experienced by the individual eccentric mass sections when the system is rotating at the excitation frequency. However, the coupling force can have any other suitable magnitude. The connection mechanism can be a mechanical connection (e.g., adhesive, clips, Velcro, etc.) with a separation force substantially equivalent to the coupling force, a magnetic connection wherein adjacent eccentric mass or mass couple sections include complimentary magnets, or any other suitable mechanism that can selectively connect adjacent eccentric mass sections together.

In one alternative, the eccentric mass 121 is collectively formed by a first and a second section (e.g., the eccentric mass 121 is divided radially into two sections), wherein the first section is a reflected duplication of the second section. In operation, the first and second sections are preferably diametrically opposed and spin about the axis of rotation of the positioning mechanism when the system vibration reaches the resonance frequency of the eccentric mass 121. In a second alternative, the eccentric mass 121 is collectively formed by a first, second, and third section with substantially the same mass, wherein the first, second and third sections are preferably substantially evenly distributed about the rotational axis when the system rotational speed reaches the resonance frequency of the eccentric mass 121. However, the eccentric mass 121 can be formed from any number of constituent sections in any suitable configuration. Alternatively, the stabilizing mechanism 700 can be any other suitable mechanism.

The FIGURES illustrate the architecture, functionality and operation of possible implementations of systems, methods and computer program products according to preferred embodiments, example configurations, and variations thereof. In this regard, each block in the flowchart or block diagrams may represent a module, segment, step, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block can occur out of the order noted in the FIGURES. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the preferred embodiments of the invention without departing from the scope of this invention defined in the following claims.

I claim:

1. A system, comprising:

- a pump, configured to statically mount to a wheel;
- a first eccentric mass rotatably mounted relative to the pump;
- a second eccentric mass rotatably mounted relative to the pump; and
- a positioning subcomponent comprising a cylindrical inner surface constraining translation of a region of the first eccentric mass element,

24

wherein the first eccentric mass is operable between:

- a pumping mode, wherein the first eccentric mass is arranged proximal to the second eccentric mass and applies a pumping force to the pump, wherein the pumping force is generated by rotation of the pump relative to the first eccentric mass; and
 - a nonpumping mode, wherein the first eccentric mass is separated from the second eccentric mass by an arcuate force applied to the first eccentric mass by the positioning subcomponent.
2. The system of claim 1, wherein the arcuate force is passively generated.
 3. The system of claim 1, wherein the arcuate force acts on the first eccentric mass after separation of the first eccentric mass and the second eccentric mass.
 4. The system of claim 3, wherein the positioning subcomponent is configured to maintain a predetermined angular position of the first eccentric mass.
 5. The method of claim 1, wherein the first eccentric mass is configured to rotate with the wheel in the nonpumping mode.
 6. The system of claim 1, wherein the first eccentric mass and the second eccentric mass are connected to a rotor and stator.
 7. The system of claim 6, wherein the rotor and stator are configured to dynamically maintain the first eccentric mass within a range of non-zero angles relative to the second eccentric mass in the nonpumping mode.
 8. The system of claim 6, wherein the second eccentric mass is connected to at least one of the rotor or the stator, wherein the rotor and stator are configured to harvest electrical energy during the pumping and nonpumping modes.
 9. The system of claim 1, wherein the first eccentric mass and the second eccentric mass are connected by coupler, the coupler applying an arcuate force to the first eccentric mass in the pumping mode.
 10. The system of claim 1, further comprising a processing system comprising a set of processors configured to control a system operation between the pumping mode and the nonpumping mode.
 11. The system of claim 1, wherein the arcuate force changes in magnitude based on an angular velocity of the first eccentric mass.
 12. The system of claim 1, wherein a center of mass of the first and second eccentric masses is offset from a wheel axis of rotation during the pumping mode and aligned with the wheel axis of rotation during the nonpumping mode.
 13. A system for pump actuation, comprising:
 - a pump, configured to statically mount to a wheel;
 - a first hanging mass and a second hanging mass, each rotatably mounted relative to the pump and operable between:
 - a pumping mode, wherein a pumping force, generated by relative motion between the second hanging mass and the pump, is applied to the pump; and
 - a nonpumping mode; and
 - a coupler mounted to the first hanging mass, wherein the coupler constrains the first hanging mass relative to the second hanging mass mode in the pumping mode, and wherein the first hanging mass and the second hanging mass transition between the pumping mode and nonpumping mode responsive to a change in a rotational speed of the wheel.
 14. The system of claim 13, further comprising a generator connected to the second hanging mass, wherein the

generator harvests energy from relative rotation between the wheel and at least one of: the first hanging mass or the second hanging mass.

15. The system of claim **13**, wherein the first hanging mass and the second hanging mass are connected to a stator and a rotor. 5

16. The system of claim **13**, wherein a centrifugal force separates the first hanging mass from the second hanging mass to transition the system from the pumping mode to the nonpumping mode. 10

17. The system of claim **13**, wherein the system is configured to transition the first hanging mass and the second hanging mass from the pumping mode to the nonpumping mode when the wheel rotation reaches a threshold velocity. 15

18. The system of claim **13**, further comprising a force mechanism coupled to the first hanging mass and configured to control motion of the first hanging mass in the nonpumping mode.

19. The system of claim **18**, wherein the force mechanism controls motion of the first hanging mass by applying an arcuate force. 20

20. The system of claim **18**, wherein the force mechanism comprises a spring.

* * * * *